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From pixels to perceptions: Capturing high-level abstract concepts in visual user-generated content

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ABSTRACT

Visual content on platforms like Instagram and TripAdvisor plays a crucial role in consumer decision-making, particularly in intangible sectors like hospitality, where consumers rely on such information to gauge service quality prior to consumption. In this study, we introduce an approach leveraging deep neural networks to identify high-level, abstract concepts in visual user-generated content (UGC) for restaurants. Given the lack of annotations on these concepts, we propose two weak labeling methods: one utilizing existing restaurant-quality signals, and the other extracting relevant labels from review texts via questionnaires. Our findings reveal that models trained on these inexpensive weak labels demonstrate moderate correlations with human judgments. Furthermore, we showcase the effectiveness of gradient-based techniques in generating visual explanations, highlighting image regions that support neural network predictions. These methods enable visual UGC analysis tasks with minimal labeling effort and allow practitioners to interpret deep neural network predictions more effectively.

1. Introduction

Today, marketers have access to a vast amount of online User-Generated Content (UGC), such as social media posts and online reviews (Babić Rosario, Sotgiu, De Valck, & Bijmolt, 2016; King, Racherla, & Bush, 2014). As a dynamic, direct, and practically limitless source of consumer opinion and feedback, UGC has revolutionized marketing research and practice and is estimated to be at least as valuable to firms as traditional market research methods (Timoshenko & Hauser, 2019). While user-generated content (UGC) exists in many modalities, the proliferation of image-centric social media networks, the widespread adoption of smartphones, and the increasingly universal availability of fast, affordable internet have made visual content (images and videos) the dominant form of UGC. For instance, more than 6.5 billion images are shared on social media platforms every day (Hartmann, Heitmann, Schamp, & Netzer, 2021). Visual content provides a rich medium for conveying complex information that text alone often cannot adequately capture and is well received as its richness pairs well with our biological predisposition towards visual information; humans process complex visual concepts rapidly and are more likely to remember items presented as pictures rather than words (Ally & Budson, 2007; Thorpe, Fize, & Marlot, 1996). As a form of UGC, visual content provides consumers with a powerful means for sharing information about products and services. However, the potential usefulness of visual forms of UGC does not come without some associated challenges in drawing inferences from a rich and unconstrained medium.

The vast majority of existing UGC research has focused on extracting insights from text-based data. In contrast, methods for analyzing visual UGC are comparatively in their early stages, with only a few studies having explored this area to date (Grewal, Gupta, & Hamilton, 2021). These early works primarily examined the effects of visual UGC, which was still in its infancy, on relevant business outcomes. More recently, studies have started to adopt a content-centric approach

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where large-scale visual content is analyzed to identify concrete, low-level visual features such as colors, shapes, patterns, or visual concepts (such as objects and scenes), associated with a business outcome. For example, Jalali and Papatla (2016) investigate the role of the color compositions of visual UGC in consumer response and interaction. More broadly, they have also been used in large-scale brand content analysis (Nanne et al., 2020). While the reliance of such studies on concrete, low-level visual features is generally justified by the fact that these features are better understood (Mkrtychian et al., 2019), existing research built upon these features often falls short of capturing the full scope of how consumers interpret and value visual UGC. Despite their widespread usage, low-level features often provide a relatively restrictive interface for modeling the higher-level semantics that are commonly associated with business-relevant dimensions that consumers value and that play a crucial role in their decision-making process. While quantifiable visual properties offer valuable insights, they do not provide a comprehensive understanding of the nuanced ways consumers use visual UGC to estimate their experiences.

High-level, abstract concepts encompass more complex and intangible concepts that consumers perceive and value in visual UGC. For example, while a user-shared photo on a review aggregation platform can naturally be seen in terms of its low-level features, the principal informative value of the image for the consumer lies in the abstract, high-level semantics within the image. These high-level, abstract concepts convey information on complex attributes like mental state (e.g., happy, sad), an evaluation (e.g., high quality, bad service), or a customer need (e.g., responsiveness). From an information-seeking perspective, consumer information needs primarily involve gathering information about relevant, visually available, high-level, abstract concepts surrounding a product or service. Consider the example of restaurants: potential customers examine visual user-generated content (UGC) to gather insights about complex concepts like food quality or the atmosphere of the restaurant. During these assessments, consumers evaluate the contents of a UGC image holistically to estimate their overall consumption experience, rather than focusing solely on individual aspects such as furniture, food, or lighting. The presence, absence, or intensity of these elements cannot be adequately described by a simple combination of specific visual properties like colors or the presence of particular objects. Atmosphere is a notable example of an abstract, perceptual concept that is critically important to the restaurant industry, and its perception relies on a complex combination of factors such as lighting, decor, and more, which are hard to precisely quantify. Furthermore, restaurants are not isolated examples of such information needs; the importance of abstract concepts is relevant to the service industry as a whole. Services are often characterized by their intangibility — being unable to be seen, felt, tasted, or touched. Given these traits and the general unavailability of pre-purchase trials, visual UGC serves as a powerful tool for consumers seeking to gain more information about the service. Thus, information on high-level abstract concepts is a vital component of consumers' information needs.

In this work, we introduce a novel methodological framework using deep convolutional neural networks to detect gradation in four high-level, abstract concepts – *food quality*, *service quality*, *value*, and *atmosphere* – within visual UGC from restaurants. Our framework addresses the challenge of obtaining visual data annotated on these attributes by proposing two novel annotation strategies. In the first proposed strategy, images within UGC are associated with a restaurant. The label assigned to this restaurant is then transferred to all associated instances. This computationally inexpensive method generates a weak supervision signal for the UGC instances using openly available annotations for restaurants. If these restaurant-level annotations are not available, the second proposed strategy employs modern text-mining methods to generate weak supervision signals, thereby facilitating the training of deep learning models without requiring extensive labeled data.

Furthermore, our approach goes beyond existing methods by focusing on detecting gradations in high-level, abstract concepts rather than simply identifying their presence or absence. We frame the detection of these concepts as a ranking task, training the model to produce an ordinal value indicating the strength of each concept. This emphasis on gradation offers valuable insights to firms by providing a nuanced, graded understanding of these concepts, which is crucial for informed decision-making.

To achieve this goal, we train an openly-available deep neural network from scratch to predict the aforementioned concepts in large-scale visual user-generated content (UGC). While commercial turnkey solutions such as Google Cloud Vision are suitable for common computer vision tasks like image labeling, face and landmark detection, optical character recognition (OCR), and tagging of explicit content, they are unsuitable for the specific task proposed in our work, as this task necessitates the training of a model tailored to a specific objective. Additionally, such services typically provide APIs that may lack transparency, reproducibility, and flexibility for training visual models on specific downstream tasks, as attempted in our work. To ensure that the resulting models are freely available, open for study, and interpretable, we opt for a fully reproducible approach built upon open models and a well-documented training recipe.

The usage of open models offers a significant additional benefit: it enables us to generate visual explanations for model predictions. For instance, in this work, we utilize Gradient-weighted Class Activation Mapping (Grad-CAM), which provides coarse localization maps indicating regions of the image crucial for a specific model prediction (Selvaraju et al., 2017). This capability addresses certain limitations stemming from the black-box nature of existing deep learning approaches and enhances transparency and interpretability for practitioners. Moreover, it equips practitioners with a valuable tool for analyzing surprising predictions and errors.

In a nutshell, our contributions are:

1. We introduce an approach for detecting the strength of high-level, abstract concepts in large-scale visual user-generated content (UGC) from restaurants using deep convolutional networks. Recognizing that collecting labels for such tasks can be prohibitively expensive, we demonstrate a method that leverages restaurant-level consumer judgments in the form of star ratings, as a weak-supervision signal for training these networks.
2. We examine the basis for predictions from the aforementioned models using a gradient-based visualization technique, Grad-CAM, which highlights regions of the image that support model predictions.
3. We propose a novel technique for mining weak-supervision labels for arbitrary concepts of interest. This technique operationalizes the neural question–answer retrieval framework by using survey questions as definitions for concepts, thereby avoiding the need for time-consuming seed-word-based approaches.

The remainder of this paper is structured as follows: Section 2 presents relevant literature from neuroscience and computer science research relevant to this work. Section 3 formally introduces the approach and technical preliminaries for training deep neural networks to capture high-level, abstract concepts in images. Section 4 introduces a novel approach for extracting labels from review texts. Section 5 details the experimental setup and provides information on data collection, model training, and evaluation. Section 6 presents results from the conducted experiments and an analysis of those results. Section 7 discusses the implications of our findings and the broader managerial insights resulting from this work, limitations and directions for future research. Finally, Section 8 concludes this work.

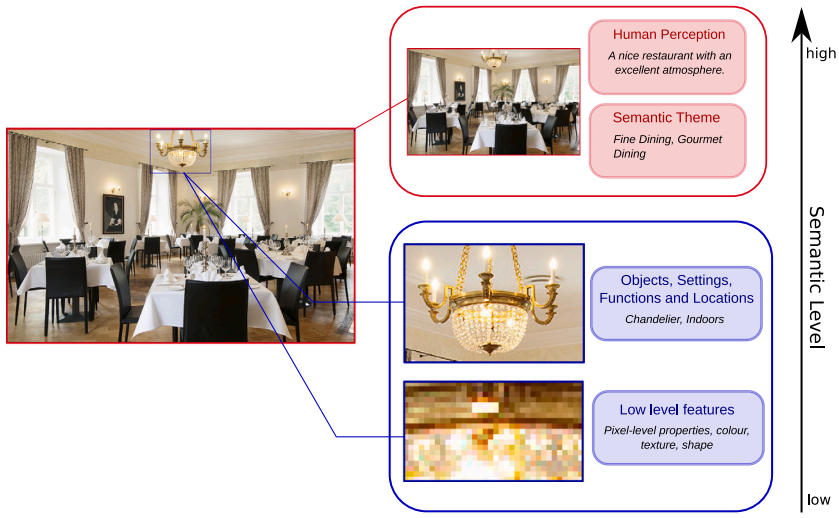

Fig. 1. Examples for various semantic levels at which an image can be analyzed by computer vision algorithms.

2. Background and related work

The exploration of high-level, abstract concepts within large-scale visual user-generated content (UGC) holds significant relevance across diverse research communities. Within information management and visual marketing, understanding these concepts is pivotal for crafting effective brand strategies and influencing consumer behavior. From a computer vision standpoint, this stream of research contributes to advancing recognition and understanding capabilities for complex properties within scenes, with applications ranging from image search to recommendation. In the realm of multimedia research, analyzing visual themes and concepts enhances techniques for content recommendation and retrieval across various media platforms. Additionally, insights from this work intersect with neuroscience, shedding light on how visual stimuli impact human cognition and decision-making processes. In the following sections, we discuss how this work builds upon existing research in these domains and provide a broad rationale for certain design choices made in this work.

2.1. Concrete and abstract concepts from a neuroscientific point of view

The distinction between abstractness and concreteness of concepts originates from significant neuroscience research in this domain. Smith (1989) define concepts as a mental representation of a class or individual, focusing on what is represented and how that information is typically utilized in categorization. In the fields of psycholinguistics and cognitive studies, concepts have traditionally been categorized into concrete and abstract types. Concrete concepts are generally associated with material objects that have clear references (e.g., house, chair), whereas abstract words are linked to non-physical entities or complex mental states (e.g., happiness, sadness), conditions (e.g., feeling), situations (e.g., encounter), and relationships (e.g., married) (Mkrtychian et al., 2019). This classification has a rich history that dates back to the work of Paivio who argued in his seminal work "Abstractness, Imagery, and Meaningfulness in Paired-Associate Learning" that concrete concepts are primarily differentiated from their abstract counterparts by their ability to evoke images more readily (Paivio, 1965). Subsequent research has built upon this classification by demonstrating the advantage of concrete semantics over abstract semantics, referred to as the "concreteness effect", demonstrating that concrete terms are learnt faster (Mestres-Missé, Münte, & Rodriguez-Fornells, 2014) and better comprehended and remembered (Schwanenflugel, Akin, & Luh, 1992; Schwanenflugel & Shoben, 1983). This effect is attributed to the richer sensory and perceptual information associated with concrete concepts, which facilitates their processing and comprehension (Mkrtychian et al., 2019).

2.2. Semantic levels of image understanding

The notion of content semantics is central to content-based image retrieval (CBIR), multimedia analytics, and computer vision tasks. Content semantics refer to the meaning and context of the visual content within an image, which principally dictate the nature and complexity of a visual learning task. This includes objects, scenes, actions, relationships, and any other semantic information that can be inferred from the image. As such, an image can carry information at multiple *semantic levels*. These semantic levels can be broadly classified as follows:

- Low-Level Features: These are basic visual features extracted directly from the image, such as color histograms, texture patterns, and edge information. While these features are essential for image representation, they do not capture the full meaning or context of the image.
- High-Level Semantics: This level involves understanding the content, context, and meaning of the image. It includes recognizing objects, scenes, actions, relationships between objects, and even abstract concepts or emotions depicted in the image. For instance, in an image of a beach with people surfing, high-level semantics would include concepts such as the beach, the surfboards, the people surfing, and the overall activity as 'surfing.'

An illustration of the varying semantic levels in an image is provided in Fig. 1. Content semantics are inextricably linked with the fundamental task of information retrieval. To retrieve images relevant to a task or query, it is essential to correctly identify the concepts, themes, and messages expressed in the image and how they relate to the information needs expressed via the query or tasks. Such needs may be relatively simple and could be addressed using well-established image processing and computer vision techniques like the detection and extraction of patterns within visual properties like color, texture, shapes, objects, actions, and events. In such tasks, traditional computer vision approaches often utilize low-level features related to color, texture, edges, and shapes that can be utilized by a computer to learn the relevant information (Chang, Sikora, & Purl, 2001; Smeulders, Worring, Santini, Gupta, & Jain, 2000). However, low-level features are often overly simplistic for complex, specialized tasks and may require a domain expert to carefully design relevant features for the task. These approaches have been largely upended by the advent of deep learning, where models utilize a combination of tightly integrated layers and end-to-end training to learn powerful task-specific features from scratch using large amounts of data (LeCun, Bengio, & Hinton, 2015). Deep neural networks designed for visual tasks commonly utilize

multiple convolutional layers — a specialized neural network building block that exploits the regular grid-like topology of images to extract visual features relevant to the task at hand LeCun and Bengio (1998). The ability to learn task-specific features from data has allowed deep learning models to be applied to varying computer vision tasks like object detection, image classification, and semantic segmentation, where they significantly outperform models built upon hand-crafted features (Ghiasi et al., 2020; Pham, Dai, Xie, Luong, & Le, 2020; Tao, Sapra, & Catanzaro, 2020).

Capturing higher-level semantics can be challenging due to the complexity and subjectivity involved in interpreting image content. Tasks centered around the detection of higher-level semantics principally suffer from the same issues that make the detection of abstract concepts difficult — the lack of concreteness and the resulting subjectivity in specifying them. Nonetheless, deep learning models have been developed for capturing the inherently abstract notions of visual memorability (Isola, Parikh, Torralba, & Oliva, 2011; Isola, Xiao, Torralba, & Oliva, 2011; Khosla, Raju, Torralba, & Oliva, 2015), popularity (Ding, Ma, & Wang, 2019; Gelli, Uricchio, Bertini, Del Bimbo, & Chang, 2015; Khosla, Das Sarma, & Hamid, 2014; McParlane, Moshfeghi, & Jose, 2014). More recently, such techniques have also been applied in the detection of complex themes like human intent (Jia et al., 2021; Wang, Yang, Zhao, Long, & Li, 2023), hateful meme detection (Lee, Cao, Fan, Jiang, & Chong, 2021; Lippe et al., 2020), and greenwashing detection (Sharma, Rudinac, Demmers, van Dolen, & Worring, 2024). These studies demonstrate the potential of deep learning models in capturing high-level, abstract concepts in visual content.

In this study, we consider concepts of interest to be abstract entities, comprising high-level semantic information that resonates with consumers. This approach is justified because consumer perceptions of these elements often involve intangible and perceptual aspects, which are inherently associated with abstract semantics. As a result, the information deemed useful to consumers tends to be predominantly abstract in nature and rarely dependent on the presence or absence of a single concrete visual component. Within the context of computer vision and multimedia research, the semantic information embedded in these attributes is typically categorized as high-level. Therefore, for the remainder of this work, we will treat the concepts under investigation as high-level, abstract entities.[5]

2.3. Computer vision for understanding the impact of visual content on business outcomes

Within the information management and marketing research domains, the examination of textual forms of UGC for analyzing business outcomes is widely established. Kumar, Kar, and Ilavarasan (2021) demonstrate this through a systematic review of the application of text mining methods, such as Sentiment Analysis, Topic Modeling, and NLP, in business management journals, identifying that social media analysis, market analysis and competitive intelligence are widely used while other themes like risk management and fake content detection are increasingly being explored. Modern natural language processing (NLP) techniques have been used to study sentiment in airline reviews and for multi-lingual opinion mining (Iddrisu, Mensah, Boafo, Yeluripati, & Kudjo, 2023; Shahade, Walse, Thakare, & Atique, 2023).

While textual analysis of user-generated content (UGC) has proven valuable in understanding user sentiment and online behavior, the landscape is rapidly evolving. As our world becomes increasingly dominated by visual communication, the way users engage with brands and products is shifting as well. This necessitates a deeper exploration of visual forms of UGC, which are now receiving significant attention in both theory and practice due to its information-rich nature and perceived trustworthiness in the eyes of consumers (Brown & Reingen, 1987; Chu & Kim, 2011; Reichelt, Sievert, & Jacob, 2014). Visual forms of user-generated content (UGC) not only carry informative effects but also persuasive ones, and there is ample evidence that visual UGC can have a substantial impact on business outcomes (Goh, Heng, & Lin, 2012). For instance, when conducting online research about restaurants, potential consumers pay attention to pictures of food and physical evidence of restaurants generated by other users (Oliveira & Casais, 2019). Similarly, on Airbnb, the presence of sellers' photos positively impacts reputation and willingness to pay (Ert, Fleischer, & Magen, 2016). In the context of hotel websites, the strategic use of pictures on the home page and throughout the website is a key element in both appeal and influence to purchase (Phelan, Christodoulidou, Countryman, & Kistner, 2011; Stringam & Gerdes, 2010). A study on review helpfulness using TripAdvisor and Yelp data found that user-provided photos combined with review texts are perceived as most helpful by other users (Ma et al., 2018). Furthermore, Zhang and Luo (2023) discovered that the volume of images shared by consumers is more predictive of restaurant survival than reviews, restaurant characteristics, competitive landscape, and macro conditions. At the content level, Jalali and Papatla (2016) examine the relationship between color compositions in UGC images and user interactions (click-rates), finding that photos with higher click-rates tend to exhibit higher proportions of green, lower proportions of red and cyan, as well as higher chroma levels of red and blue.

While visual content can have a substantial effect on business outcomes and consumer perception of a product or service, its analysis is particularly challenging for offerings with intangible elements, as they are not easily captured in lower-level visual properties, such as colors, textures, and the presence of specific objects. For such tasks, deep learning-based approaches are increasingly gaining popularity. Recent studies focused on content analysis have leveraged existing computational techniques to examine vast collections of user-generated content (UGC) and identify visual properties correlated with desirable business outcomes. Specialized deep neural network architectures can automatically extract relevant patterns underlying a business outcome, such as consumer engagement or visual appeal, from visual content. These approaches excel in scenarios where the volume of visual content is too large for manual analysis.

For instance, Nanne et al. (2020) employed computer vision to analyze large-scale brand content by extracting objects present in brand-related social media images using scientific and commercial object detection models. Similarly, Rietveld et al. (2020) analyzed a vast amount of visual UGC on Instagram to investigate the effects of visually-induced emotional arousal on customer engagement.

More closely related to our work, Liu et al. (2020) trained a convolutional deep neural network to detect the presence of non-functional brand attributes, including 'ruggedness' and 'glamour,' in brand-related content on Instagram. These studies critically rely on the ability of deep learning approaches to automatically learn complex patterns between images and business outcomes from large-scale training data. An overview of literature focusing on content-based analysis, including the modalities explored, research objectives, and methodological frameworks, is provided in Table 1. While these studies have made significant strides in understanding the impact of visual content on business outcomes, they primarily focus on concrete, low-level visual properties. In contrast, our work aims to capture high-level, abstract concepts in visual UGC, which are more closely aligned with consumer perceptions and decision-making processes.

3. Detecting high-level, abstract concepts in restaurant images with deep neural networks

In this work, we present an approach that uses deep convolutional neural networks, trained in a weakly-supervised fashion, to detect

[5] All concepts examined in this work are predominantly high-level and abstract in nature. Thus, for the sake of brevity, we will often refer to them simply as "concepts", implying their inherent high-level and abstract characteristics.

Table 1
A review of selected literature that performs content-based analysis of visual UGC and social media data. Our work presents the first step towards using large-scale data for training deep learning models in a *weakly-supervised* manner.

Study	Modality		Target Variable	Methodology
	Image	Text		
Jalali and Papatla (2016)	✓		Click-Rates	Beta-Regression using MCMC
Lee, Hosanagar, and Nair (2018)	✓		Engagement (Likes/Comments)	Correlation Analysis
Ma, Xiang, Du, and Fan (2018)	✓	✓	Review Helpfulness	Supervised Classification
Nanne et al. (2020)	✓		Brand Attributes	Clustering, Supervised Classification
Rietveld, van Dolen, Mazloom, and Worring (2020)	✓	✓	Engagement (Likes/Comments)	Supervised Classification, Poisson Regression
Liu, Dzyabura, and Mizik (2020)	✓		Perceptual Brand Attributes	Supervised Classification
OURS	✓		High-Level, Abstract Attributes	Weakly Supervised-Ordinal Regression

the strength of high-level, abstract concepts in large-scale visual user-generated content (UGC). This represents the essence of our work. As such, this task contains two critical design choices: (1) the use of deep convolutional networks as the visual inference backbone and (2) the use of weakly-annotated data for training such networks for our task. The use of deep neural networks as a visual inference backbone is guided by the understanding that it is often difficult to manually identify visual features underlying the perception of high-level, abstract concepts. Images contain information at various levels of abstraction. At the primitive level, images are built from regular grid-like pixel arrangements. Specific pixel arrangements give rise to structures like edges, textures, shapes, and objects. Multiple objects placed in a specific spatial configuration give rise to scenes. At the highest level, perceptual properties of the scene, such as memorability, visual sentiment, and interestingness, emerge from a complex interplay between properties at lower levels. Each stage represents a *semantic level* encoding specific information about the image (Smeulders et al., 2000). Lower semantic levels capture objective properties of the scene, whereas high semantic levels capture elements of the image often associated with abstract human interpretations. Since high-level, abstract concepts are inextricably linked with human interpretation, visual features underlying such concepts may involve a complex interplay between image features defined at different semantic levels. Deep neural networks are well-suited for capturing such complex relationships between image features and high-level, abstract concepts, as they can automatically learn task-specific features from data (LeCun et al., 2015).

Computer vision models traditionally handle tasks in two stages. In the first stage, images are converted into features — salient components of the image that contain information relevant to the prediction model. Traditional computer-vision approaches utilize relatively simpler, general-purpose features, including statistics on colors, geometric primitives like key points, edges, and textures, or the shape and geometry of different regions within the image. Features relevant to a task are selected through a feature engineering process. In the second stage, the visual representation obtained from selected features is used as an input for machine learning models relevant to the task at hand.

While traditional computer-vision approaches are fairly simple and generally interpretable, feature engineering is often a time-consuming process that relies strongly on domain expertise and becomes increasingly difficult and brittle for complex tasks. Deep learning automates the feature engineering process by allowing task-relevant features to be systematically discovered from large amounts of training data (Chollet, 2017). At a structural level, a neural network is a function f mapping an input $\mathbf{x}$ to an output y. It is built from a composition of multiple simpler functions or layers such that $f = h_1 \circ h_2 \circ h_3 \ldots \circ h_n$, where h_n is the nth layer. A layer h_i transforms the input $\mathbf{x}$ such that $y_i = h_i(\mathbf{x})$. The output y_i serves as the input for the subsequent layer h_{i+1}.

Convolutional neural networks further assume that the input $\mathbf{x}$ has a spatial, grid-like structure such that $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, where H and W are the height and width (spatial dimensions) of the input volume, and C is the size of the activation volume, often referred to as channels. Channels are layers of primary colors that comprise a color image; a common example is RGB (Red, Green, and Blue) channels. Convolutional neural networks exploit strong spatially-local correlations in images to constrain connectivity within neurons. Within a deep convolutional neural network, convolutional filters repeatedly extract and refine visual properties relevant to a task into an optimal representation for a prediction model. This representation serves as the input for the prediction model in the final layer(s) of the network that transforms it into the desired output.

This strategy of coupling and jointly optimizing the feature extraction and prediction stages often leads to better performance, and deep convolutional neural networks have enabled unprecedented breakthroughs in computer vision tasks like object detection, image classification, semantic segmentation, and image captioning (LeCun et al., 2015).

Since a priori assumptions about which visual features contribute to better perceptions of high-level, abstract concepts cannot be made, we rely on deep neural networks to discover these features from training data by composing information at multiple semantic levels into task-relevant representations. As shown in Khosla et al. (2014), layers within deep neural networks capture information at various levels of abstraction. Early layers within the network capture image properties like edges and colors, intermediate layers detect mid-level properties like textures, shapes, and objects, and final layers capture higher-order scene-level properties. A combination of hierarchical features can be used to build powerful representations (Zeiler & Fergus, 2014).

We formulate the task of learning the strength of a high-level, abstract concept expressed within an image as a supervised learning task where a deep neural network is trained to map an input image I to a numerical value Y indicating the relative strength of the concept. A supervised learning task formulation requires (1) a labeled image

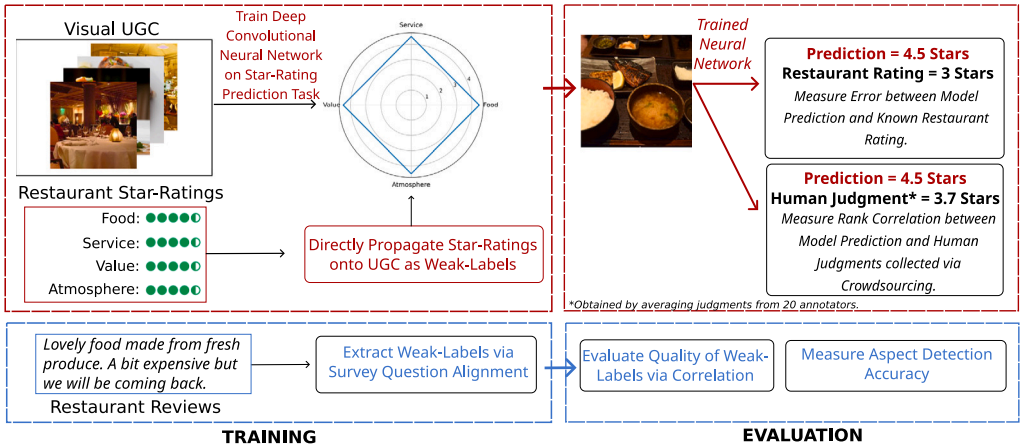

Fig. 2. Tripadvisor pages for restaurants are collected and processed to extract all images, restaurant star ratings on "food", "service", "value" and "atmosphere", and all text reviews. Star ratings are directly propagated to a restaurant's images as a weak label on the aforementioned concept and used for training a deep neural network to predict these concept for unseen images. Alternatively, weak labels for arbitrary high-level, abstract concepts of interest can be mined from text reviews using a novel text-mining approach.

dataset where each image is annotated with a score denoting the relative strength of the concept under examination, and (2) an objective function that allows neural networks to learn the relationship between the input images and the labels. A visual overview of the approach is available in Fig. 2.

It is worthwhile to note here that despite the simplicity, ease of use and general availability of Software-as-a-Service (SaaS) AI platforms, such as Google Cloud Vision and Amazon Web Services, the task under consideration in this work essentially rules out their usage. The aforementioned platforms often allow access with the service through tightly-controlled APIs that often make it difficult to modify the model on a specific downstream task. Our approach of training an open-model locally allows greater flexibility, reproducibility and transparency with the added benefit of allowing the usage of interpretability techniques that rely on access to model internals.

3.1. Utilizing restaurant quality signals as aggregated, weak supervision

The notion of weak supervision – the use of limited, noisy, or imprecise information as supervision signals – is central to this work. In essence, this is a real-world limitation, as visual user-generated content (UGC) "in-the-wild" often lacks meaningful instance-level annotations due to its unstructured nature.

To generate labels for visual UGC without resorting to expensive manual labeling, we propose a labeling approach that associates images with a set of *central entities* that are already labeled. For example, while individual images of a restaurant on Tripadvisor[6] may not be annotated with concepts like food quality or restaurant atmosphere, the restaurant often contains aggregated star-ratings collected from multiple reviews. Our first approach obtains weak labels by propagating information associated with a restaurant onto its visual UGC. This indirect labeling approach is especially suitable for cases where the labels of interest are already available on content sharing or social networking websites.

Formally, given a set of restaurants R, a set of images I, and a set of numerical labels for an arbitrary high-level, abstract concept Y, two associations are known. The first association links images with the restaurants, $D_1 = \{(i_j, r_j)\}$ where $i_j \in I$ denotes the jth image and $r_j \in R$ denotes the restaurant to which it belongs. The second association links restaurants with a numerical label (e.g., star-ratings, review volume, etc.), $D_2 = \{(r_j, y_j)\}$ where $r_j \in R$ denotes the jth restaurant and $y_j \in Y$ denotes the corresponding label associated with the restaurant. Using the set of restaurants R, a weakly-labeled dataset D_W can be constructed by copying the numerical label for a restaurant onto every image associated with that restaurant, such that $D_W = \{(i_j, y_j)\}$ where i_j denotes the jth image and y_j denotes the corresponding numerical label on an arbitrary high-level, abstract concept. By using restaurants as the link, we can associate labels Y with the originally unlabeled image-set I. Thus, D_W allows us to transform the problem of analyzing the image-set I into a weakly supervised learning task. In this formalization, the numerical label associated with a restaurant serves as a label for all images belonging to that restaurant. For the rest of this work, we will refer to indirect labels acquired in this fashion as weak labels.

It is worth noting that our proposed approach and weak labels in general offer a pragmatic compromise that allows labels to be obtained easily and cheaply at the cost of increased label noise and decreased label diversity. Despite their noisy, coarse, and generally imperfect nature, weak labels can still be used for building relatively strong predictive models. Supervision in the form of weak and partial labels has been effectively utilized for computer vision tasks like semantic segmentation (Papandreou, Chen, Murphy, & Yuille, 2015; Xu, Schwing, & Urtasun, 2015) and, perhaps closer to our work, in neural ranking models (Dehghani, Zamani, Severyn, Kamps, & Croft, 2017). In the context of our work, weak labels are particularly useful for training deep neural networks to predict high-level, abstract concepts in visual UGC, as they provide a computationally inexpensive way to obtain labels for a large number of images.

3.2. Detection of high-level, abstract concepts as weakly-supervised star-rating prediction task

The availability of D_W enables us to formulate the problem as a (weakly) supervised learning task. Since our objective is to learn gradations in high-level, abstract concepts, the labels y must assume a graded, ordinal form. In this work, we utilize star ratings — a rating scale commonly used in the hospitality and entertainment industry for rating restaurants, hotels, and movies on concepts relevant to the offering. At a conceptual level, star ratings are a discrete and ordinal source of information that can be seen as ranks. Formally, a rating scale with N distinct star-ratings can be seen as a set of N distinct ranks $\mathcal{R} = \{r_1, r_2, \ldots, r_N\}$ such that $r_1 \prec r_2 \prec \ldots \prec r_N$. The problem of

[6] https://www.tripadvisor.com/

detecting restaurant concepts from images can then be formalized as a learning task where, given an image $\mathcal{I}$, labels $\mathcal{Y} \in \mathcal{R} = \{r_1, r_2, \dots, r_N\}$, the task is to learn an optimal ranking function $f : \mathcal{I} \rightarrow \mathcal{Y}$. Since our task is to predict ordered star-rating categories, it can be viewed as an ordinal regression task. Ordinal regression is a statistical method used to model relationships between predictor variables and an ordinal outcome variable, which has ordered categories but not necessarily equal intervals between them. It is particularly well-suited for predicting star ratings for restaurants because star ratings typically fall into ordered categories (e.g., ★, ★★, ★★★, etc.), and the intervals between these categories may not represent equal differences in quality.

Earlier works have formulated the prediction of ordinal outputs as a multi-class classification problem, disregarding the natural ordering of the target categories (Levi & Hassncer, 2015; Rothe, Timofte, & Van Gool, 2015). However, this approach often results in the loss of information contained in the ordering of the target categories. Recently, specialized machine learning approaches have been proposed for ordinal regression that can incorporate the information contained in the natural ordering of ranks into the prediction model. These approaches have been shown to outperform traditional multi-class classification models in ordinal regression tasks (Diaz & Marathe, 2019; Rothe et al., 2015).

In this work, we utilize an ordinal regression framework that transforms the ordinal regression task into a multi-class classification task using a soft target encoding scheme that integrates metric penalties into the label representations. This approach allows us to leverage the ordinal nature of the target categories and the information contained in the ordering of the ranks for training the deep neural network (Diaz & Marathe, 2019). This approach converts an ordinal regression task into a classification task using a soft target encoding scheme that integrates metric penalties into the label representations. In a nutshell, each rank is transformed into an inverted-V-shaped probability distribution over the set of ranks, with the highest probability assigned to the true rank r_T. The probability for a rank r_i is inversely proportional to the distance between r_i and the true rank r_T. Consequently, the probability associated with a rank decreases as its distance from the true class increases, serving as a measure of the degree of misclassification. Unlike multi-class classification, which treats all misclassifications as equally incorrect, this ordinal regression framework imposes a variable misclassification penalty that rises with the magnitude of misclassification, indicated by the *distance* between the predicted class and the true class.

For instance, when the true class is 5, predicting 4 incurs a smaller misclassification penalty than predicting 2, although both predictions are incorrect. In this work, we denote this approach as "OR-STE" (Ordinal Regression with Soft Target Encoding). For each rank, this transformation yields an inverted-V-shaped probability distribution, as demonstrated in Fig. 3.

Formally, given a set of ranks $\mathcal{R} = \{r_1, r_2, \dots, r_N\}$, a true rank $r_T \in \mathcal{R}$, and an appropriate distance measure for ranks ϕ, the distribution over the set of star-ratings is defined by Eq. (1). Since ordinal regression models are generally trained to minimize the mean absolute error between predicted and true ranks, we set the distance measure ϕ as the absolute error between ranks, which is given by $\phi(r_i, r_T) = |r_i - r_T|$. The probability distribution over the set of ranks $\mathcal{R}$ is then computed using the softmax function, as shown in Eq. (1).

$$P(r_i, r_T) = \frac{\exp(-\phi(r_i, r_T))}{\sum_{n=1}^{N} \exp(-\phi(r_n, r_T))} \quad \forall r_i \in \mathcal{R} \tag{1}$$

At a high level, the model transforms an input image into a set of unnormalized outputs over the set of ranks $\mathcal{R} = \{r_1, \dots, r_N\}$. These outputs are then further transformed by a normalized exponential (softmax) function into a probability distribution Q over the set of N output ranks, where Q_i denotes the estimated probability for rank r_i. The Kullback–Leibler Divergence is subsequently used as the measure of the difference between the target distribution P and the predicted distribution Q, as specified in Eq. (2). The complete training algorithm is provided in Algorithm 1.

$$D_{KL}(P \parallel Q) = \sum_{r \in \{r_1, \dots, r_N\}} P(r) \log \left(\frac{P(r)}{Q(r)} \right) \tag{2}$$

This formulation has several advantages. First, it transforms an ordinal regression task into a well-studied multi-class classification task, which means that no structural modifications to a pre-trained deep neural network are required beyond replacing the final layer (classifier). More importantly, this structural similarity allows for the direct application of interpretability methods originally designed for multi-class classification, such as Grad-CAM (Selvaraju et al., 2017), without modification. Furthermore, since a classifier emits a probability distribution over a set of outputs (ordinal ranks), the expected value of this output probability distribution can be utilized as a scalar measure encoding the weighted average of the output distribution. This measure, formalized in Eq. (3), provides a point estimate for the model's output and can be used for ranking images.

$$\mathbb{E}(Q) = \sum_{n=0}^{N} Q_n * n \tag{3}$$

Algorithm 1: Algorithm for the Weakly-Supervised Learning Task

Data: $R \leftarrow$ set of restaurants.
$I_r \leftarrow$ set of images for a restaurant.
$y \in \{r_1, \dots, r_N\} \leftarrow$ score for restaurant on a concept.

Function `TRAIN BATCH`(nn, $\mathbf{x}$, y):
  for $b = 1, 2, \dots, B$ **do**
    $Q_b \leftarrow nn(\mathbf{x}_b)$ ▷ nn = `deep neural network`
  end
  $loss \leftarrow 0$
  for $b = 1, 2, \dots, B$ **do**
    $P_b \leftarrow f(y_b, \{r_1, r_2, \dots, r_N\})$ ▷ `Equation (1)`
    $loss \leftarrow loss + \mathcal{L}_{KLD}(P_b \parallel Q_b)$
    $nn \leftarrow$ update parameters with backpropagation
  end

3.3. Class activation mappings

Deep neural networks often trade off model simplicity for predictive power, which can limit practitioners' ability to analyze their inner workings. However, the ability to inspect the decision-making process in deep learning can be a powerful resource for practitioners trying to diagnose problems or understand the rationale behind a model's predictions. In this work, we utilize Class-Activation Mapping (CAM) as a tool for identifying salient regions in an image that support a prediction (Zhou, Khosla, Lapedriza, Oliva, & Torralba, 2016). Specifically, we employ Grad-CAM, a generalization of the CAM framework that leverages gradients from the final convolutional layer of a deep neural network to compute the importance of a visual region towards a specific classifier output (Selvaraju et al., 2017). This involves computing weights for each neuron in the convolutional layer towards a specific model prediction. These weights are then multiplied with the convolutional layer activations to yield a 2D heatmap that indicates salient regions within the image that support a specific prediction.[7] The obtained heatmaps can be overlaid on the input image to generate visual explanations for a prediction. Within our study, Grad-CAM is used to generate visual explanations by highlighting regions of the image that support a specific rating for a high-level, abstract construct.

[7] c.f. Selvaraju et al. (2017) for more information on how weights are computed.

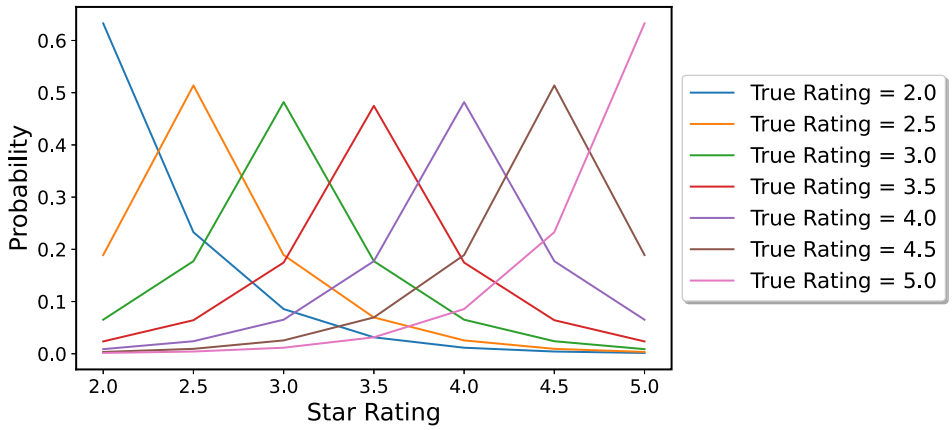

Fig. 3. Probabilistic Labels generated for the "Soft-Target Encoding" approach inspired by Diaz and Marathe (2019). For each curve, the probability (y-axis) peaks when the predicted rank matches the true rank and progressively decreases with increasing distance from the true rank. The distance measure (ϕ) used for these plots is absolute error $\phi(r_i, r_T) = |r_i - r_T|$.

4. Extracting weak supervision from review text

The approach presented in the preceding section utilizes star ratings on a set of high-level, abstract concepts as a weak supervision signal for images. However, in the absence of such ratings, the proposed approach cannot be used. For instance, consider the example of Tripadvisor, which only collects and displays star ratings on food, service, value, and atmosphere; a practitioner attempting to train a model for detecting "restaurant cleanliness" in images will find that information on this concept is neither collected nor displayed by the UGC platform. In this section, we introduce an unsupervised text-mining approach for extracting a supervisory signal on any abstract concept of interest. This approach extracts weak labels by identifying whether an opinion on a concept is expressed in a review sentence and then computing the sentiment of the expressed opinion by analyzing the sentiment of the sentence that expresses it. This is achieved through a novel operationalization of the Question–Answer retrieval task on Review Texts – a rich, topically unconstrained source of information. This approach brings together question–answer retrieval and sentiment detection and is closely related to Aspect-based Sentiment Analysis, where the objective is to capture the fine-grained sentiment associated with an aspect in a given text (Pontiki et al., 2015). While existing approaches for the aspect-based sentiment analysis task can be used for generating weak labels, they require heavy supervision and can only predict sentiment on the limited set of aspects they were trained upon. In practice, this will require practitioners to invest significant efforts in preparing a reasonably large dataset of review sentences annotated with consumer sentiment on the concept of interest. As with any annotation effort, this is expensive and time-consuming. With the proposed approach, we hope to offer a simple way to generate weak labels without extensive effort on the part of the annotator. Within the context of this work, this approach presents an alternative strategy for generating weak labels if a more direct form of weak labeling, like the label propagation approach discussed in the prior section, is infeasible. The proposed approach is detailed in the following sections.

4.1. Approach overview

Reviews are a powerful source of information, especially for experiential products (Zhao, Yang, Narayan, & Zhao, 2013), and have been used as a rich medium for mining consumer satisfaction (Tirunillai & Tellis, 2014). Review texts, in particular, provide a rich and unstructured pool of consumer opinions on a wide variety of relevant attributes. In the proposed approach, each sentence within a review is considered an expression of the consumer's satisfaction (or dissatisfaction) with a specific attribute or aspect of their experience. For example, the review sentence – *"the food was bland"* – expresses displeasure with the food. More specifically, each sentence within a review can be seen as an opinion expressing a "sentiment" on a specific concept.[8] At a high level, the proposed approach seeks to identify the concept discussed in the sentence and the general sentiment expressed within the sentence separately. By aggregating consumer sentiments on various concepts over multiple sentences, each review can be transformed into a numerical signal capturing a consumer's satisfaction (or dissatisfaction) with concepts of interest. This involves two subtasks: (1) detecting which concept(s) are discussed in a sentence, and (2) detecting the sentiment associated with that sentence. The first subtask is operationalized as a question–answer retrieval task, where the objective is to identify the survey question that is best answered by a review sentence. The second subtask is operationalized as a sentiment detection task, where the objective is to detect the sentiment of a review sentence. The output of these subtasks is then aggregated over multiple sentences to generate a weak label for the concept of interest. The complete approach is detailed in the following sections.

4.2. Concept and sentiment detection

The first step in our approach requires detecting the concept discussed in a sentence. We propose a novel utilization of a long-standing marketing research tool – survey questionnaires – as a means for extracting the concept discussed in a sentence. Traditionally, questionnaires are used to collect and examine consumer opinions on topics of interest. These questionnaires typically consist of questions framed as statements that consumers can agree or disagree with to varying extents. We propose an inverted view of this setup, where every sentence in a review is treated as a noisy response to a survey question. We use survey questions as an information-retrieval tool for capturing the concept discussed in a review sentence. For example, the review sentence *"The food was bland"* can be seen as a response to a survey

[8] While the aspect-based sentiment analysis community refers to topics as "aspects", we continue to use the term "concepts" to keep them in line with the other contributions presented in this work.

query *"Was the quality of the food satisfactory?"*. By associating a review sentence with a survey question, we can further discover the concept expressed in the sentence. Since it is known that the question *"Was the quality of the food satisfactory?"* relates to food quality, we can infer that the original review sentence discusses food quality.

Computationally, we operationalize concept detection as a question retrieval task where, given a review sentence (answer) and a set of survey questions, the objective is to discover the survey question that is best answered by the review sentence. We chose a question–answer retrieval setup because achieving similar results with seed-word-based approaches would require large word lists containing commonly used words, synonyms, and common misspellings of such terms. In contrast, the number of survey questions required for this task is dramatically lower, as question–answer models are relatively resilient to small changes in framing and structure of responses.

For each review sentence, we compute the inner product of its vector representation with vector representations for all survey questions. For this task, we use a variant of the Universal Sentence Encoder model trained on the question–answer retrieval task (Cer et al., 2018; Yang et al., 2019). The scalars produced by the inner product between a survey question and a review sentence denote the numerical likelihood that the review sentence provides an answer to the survey question. The survey question yielding the highest inner product with a review sentence is considered the most relevant survey question for that sentence.

The second step is to ascertain the sentiment of the sentence, which can be achieved using off-the-shelf sentiment detection models. Specifically, we use the Valence Aware Dictionary and Sentiment Reasoner (VADER), a rule-based sentiment analysis model that is specifically tuned for social-media text, to detect the sentiment of review sentences (Hutto & Gilbert, 2014). VADER generates a score on a continuous scale from −1 (extremely negative) through 0 (neutral) to +1 (extremely positive). For a review consisting of multiple sentences, detected concepts and the corresponding sentiment on those concepts can be averaged over multiple sentences.

Algorithm 2: Algorithm for Extracting Weak Supervision from Review Text

Data: $R \leftarrow$ Review text.
$R_i \leftarrow$ ith sentence of the review.
$C \leftarrow \{c \mid c$ is a concept of interest$\}$
$D_c \leftarrow \{d_c \mid d_c$ is a survey question for concept c$\}$
$|S| \leftarrow$ Size/Cardinality of an arbitrary set S
$relevance(R_i, d_{c,j}) \leftarrow$ relevance measure encoding the suitability of review sentence R_i as a response to the j_{th} survey question under concept c.

Function COMPUTE REVIEW CONCEPT ALIGNMENT(R, C, D_C):
  for $i = 1, 2, \ldots, |R|$ **do**
    $sentiment_i \leftarrow$ VADER(R_i)
    for $c = 1, 2, \ldots, |C|$ **do**
      for $j = 1, 2, \ldots, |D_c|$ **do**
        $rel\{R_i, d_{c,j}\} \leftarrow relevance(R_i, d_{c,j})$
      end
    end
  end
  for $i = 1, 2, \ldots, |R|$ **do**
    $\mathbf{a}_i \leftarrow [rel(R_i, d_{1,1}), rel(R_i, d_{1,2}), \ldots]$
    $\mathbf{b}_i \leftarrow \text{argsort}(a_i)[-k:]$ ▷ only retain top-k detections
    $\mathbf{z}_i \leftarrow \mathbf{a}_i[\mathbf{b}_i] * sentiment_i$
    $\mathbf{y}_i \leftarrow$ MEANPOOL SENTIMENT FOR CONCEPT(z_i)
  end
  $\mathbf{y} = \sum_{i=1}^{|R|} \mathbf{y}_i$

At an algorithmic level, we compute the sentiment for a long review by aggregating the sentiments expressed in the constituent sentences. Each review is first broken down into sentences. A representation vector for this sentence and the survey questions is then obtained from the Universal Sentence Encoder. We then compute the inner product of the sentence vector with every survey question vector. While the highest inner product should, in principle, correctly identify the most relevant survey question answered by the sentence, this is often not the case. We observed that the matching process is often noisy and sensitive to framing and structure of review sentences, and reliance on a single survey question for each attribute can lead to a brittle model. To alleviate these issues, we associate multiple survey questions with each concept of interest. The rationale behind this is similar to the choice of K in K-nearest neighbors classification – a review sentence will be linked to more correct neighbors (survey questions related to the concept discussed in the sentence) than incorrect neighbors (survey questions related to concepts NOT discussed in the sentence). Thus, for each review sentence, we compute the top-K most relevant survey questions. We then compute the sentiment associated with the review sentence and attribute it to the top-K most relevant survey questions. In the final stage, we perform two aggregation steps. We first aggregate sentiments over multiple review sentences into a single aggregated review vector. This is followed by an aggregation of sentiment over the multiple survey questions for each attribute into a single sentiment for each attribute. A demonstrative overview of this approach, using one survey question per attribute, is provided in Fig. 4 and an overview of the algorithm is provided in Algorithm 2.

5. Experimental setup

This section outlines the key steps involved in preparing our data for analysis. We describe the data collection process, followed by the pre-processing steps necessary to ensure data quality and consistency. Additionally, we introduce the baseline and reference models that serve as a foundation for our research. Finally, we provide an overview of the model training procedure, including the evaluation measures used to assess performance.

5.1. Data collection and preprocessing

The data used in this study was collected from Tripadvisor, a travel website that allows consumers to post text reviews and images for restaurants and hotels. For restaurants, Tripadvisor enables consumers to provide feedback on a star-rating scale from 1 to 5 stars across five components of their dining experience: an overall rating, as well as optional ratings on food, service, value, and atmosphere. Additionally, the platform requires consumers to provide a text review. For each restaurant, Tripadvisor internally aggregates individual ratings on these aspects to provide consumers with a summary rating. This summary rating is displayed on the restaurant's page and is used to rank restaurants on the platform. The availability of these ratings on the platform makes Tripadvisor an ideal source for our study, as it provides a rich source of information on consumer opinions and preferences.

In this work, we analyze restaurants within a single city to allow for a more direct comparison, as they operate within similar local contexts regarding regulations, consumer preferences, and market dynamics. We selected Amsterdam, The Netherlands due to its exceptional diversity in culinary offerings, restaurant designs, aesthetics, and price ranges. This choice ensures that our dataset encompasses a broad spectrum of these factors, maximizing the visual richness and diversity of our dataset. For our study, we gathered Tripadvisor data from 3362 restaurants in Amsterdam. We gathered images from the gallery page and text reviews by collecting text from multiple review pages for each restaurant. During the processing stage, we removed restaurants with fewer than 20 reviews from our dataset, as ratings associated with such restaurants may be unreliable. Furthermore, we converted all restaurants with ratings less than two stars to handle the low number of samples for ratings lower than two stars. These processing steps resulted in a dataset with 1726 restaurants.

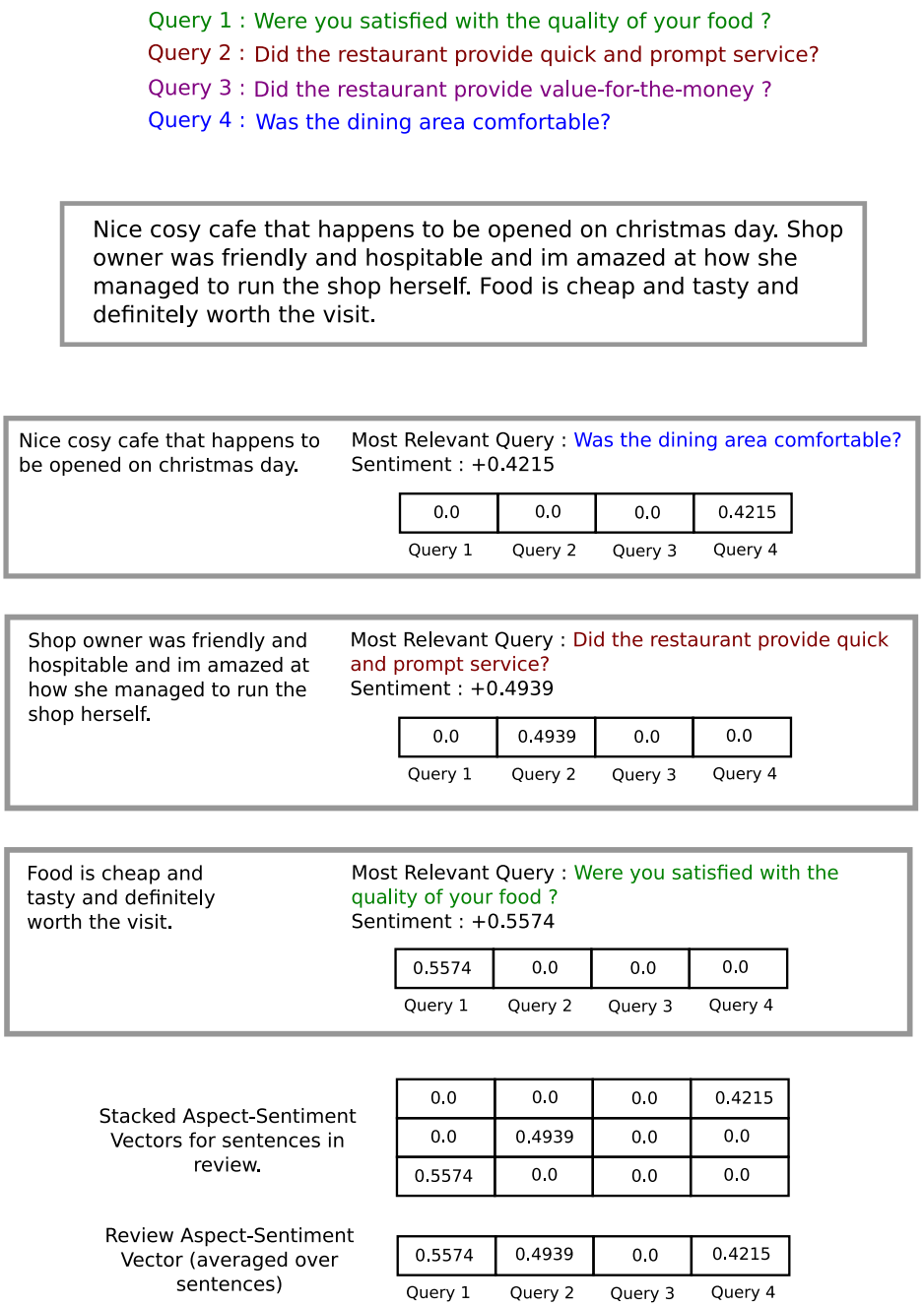

Fig. 4. The diagram demonstrates the label mining approach in a simplified manner, where each review sentence is matched with a single survey question.

We then split the restaurants into a "training+validation" set (80% of total restaurants) and a "test" set (20% of total restaurants) using an 80:20 ratio. The "training+validation" set was further divided using a 90:10 ratio to obtain a "training" set (approximately 72% of the total restaurants) and "validation" set (approximately 8% of the total restaurants). Since the data is split into restaurants rather than images, this splitting approach aligns with the commonly applied group cross-validation strategy. Moreover, since a restaurant can only be present in one split, our splitting strategy prevents potential information leakage between training and test sets. Restaurants within splits were randomly chosen, and no stratification was applied. Statistics for the number of restaurants and images in the resulting splits are provided in Table 2.

5.2. Model architecture and task formulation

For each concept, we train three models. The first model corresponds to an ordinal regression approach that utilizes a soft-target encoding scheme inspired by Diaz and Marathe (2019). We refer to this approach as OR-STE (Ordinal Regression with Soft Target Encoding). To provide a comparative baseline for the OR-STE approach, we also train all models on an alternative ordinal regression approach

Table 2
Number of restaurants and images in training, validation, and test splits.

Split	Restaurants	Images
Total	1726	151,176
Training	1242	107,276
Validation	138	13,663
Test	346	30,237

introduced in Cao, Mirjalili, and Raschka (2020), which transforms an ordinal regression task into an extended binary classification task. In this approach, each classifier predicts whether the input exceeds a specific rank, and outputs from individual binary classifiers are aggregated to determine the predicted rank. This approach is designated as OR-EBC (Ordinal Regression with Extended Binary Classification) and serves as a contemporary ordinal regression baseline for the OR-STE approach. Finally, we train a multi-category classification model using categorical cross-entropy loss as a naive baseline to illustrate performance degradation in models that ignore ordinal relationships between ranks by treating them as nominal classes. This model uses the notation CLS-CE (Classification with Cross-Entropy loss).

A pretrained version of ResNet-50, a well-studied modern deep convolutional neural network that has been shown to work well on a wide variety of visual inference tasks (He, Zhang, Ren, & Sun, 2016), serves as the visual inference backbone for our work. Specifically, the ResNet-50 model is pre-trained on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC; Deng et al. 2009, Russakovsky et al. 2015) and fine-tuned on our dataset by replacing the final classification layer with a task-specific version. For OR-STE and CLS-CE approaches, the final ImageNet-specific linear classifier with 1000 output classes is replaced with a new classifier with 7-output classes corresponding to 7-star ratings (2.0–5.0) followed by a 7-way softmax. For OR-EBC, the outputs share the same weight parameters but have different bias parameters to guarantee binary classifier consistency (Cao et al., 2020). In all task formulations, the ILSVRC parameters can thus be seen as parameter initialization for the ResNet-50 backbone (except for the replaced final classifier). This approach, formally known as transfer learning, adapts the information stored in a trained deep learning model for a different but related task and is often desirable when there is insufficient well-annotated data to train a model from scratch (Pan & Yang, 2009). For the interested reader, a model diagram for ResNet-50 is provided in Appendix B.

5.3. Model training

All models are trained for 100 epochs using stochastic gradient descent with a momentum factor of 0.9. During training, we employ multiple image transformations as data augmentation methods to improve generalization and prevent model overfitting. The optimal model is chosen by minimizing the mean absolute error on the validation set. Through empirical testing, we determine an optimal learning rate of 1×10^{-4} and a batch size of 64. All models are implemented using PyTorch (v1.12) and trained with the PyTorch Lightning framework (v1.6.4) on a machine equipped with 12 CPU cores and a single NVIDIA GeForce GTX 1080 Ti GPU.

5.4. Model evaluation

The models presented in this work are evaluated using two types of labels: (1) weak labels from a held-out test set, and (2) human judgments obtained through a crowdsourcing experiment conducted on Amazon Mechanical Turk.

5.4.1. Evaluation with weak labels

For evaluating models using the available weak-supervision sources, we use weak labels (star ratings) propagated from the restaurants onto images on a held-out test set. For these labels, we compute three standard measures that are commonly used for evaluating ordinal regression models — *Mean Absolute Error (MAE)*, *Mean Squared Error (MSE)* and *Root Mean Squared Error (RMSE)*.

$$MAE(y,\ \hat{y}) = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{4}$$

$$MSE(y,\ \hat{y}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{5}$$

$$RMSE(y,\ \hat{y}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}, \tag{6}$$

For evaluating models using the available weak-supervision sources, we utilize weak labels (star ratings) propagated from the restaurants onto images in a held-out test set. To assess model performance, we compute three standard measures commonly used for evaluating ordinal regression models: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Here, y denotes the ground-truth rank, and $\hat{y}$ represents the rank predicted by the model. Since this evaluation approach employs test labels propagated from restaurants to images in the same manner as training labels, these measures can reliably indicate overfitting and a lack of generalization if a large disparity is observed between these measures on the training set and the validation set.

5.4.2. Model evaluation with human judgements

The second form of evaluation in this work relies on strong image-level judgments collected through crowd-sourcing to measure whether model predictions align with human perception. To this end, we designed a crowd-sourcing experiment where independent judges were asked to evaluate a set of images on the four high-level, abstract concepts studied in this work. We selected 656 images for annotation by sampling a fixed number of images for each star rating within each concept. Manual selection of images was deemed unsuitable, as it could have inadvertently transferred the author's biases onto the annotators via the selection process. Random sampling was also considered unsuitable, as the original dataset is imbalanced and random sampling would replicate the data distribution of the test set (cf. Fig. 5) in the annotation set, further exacerbating the lack of representation of images associated with infrequent star-ratings. Instead, for each concept and available star rating, approximately 150 images were randomly selected for inclusion in the annotation set. For each image, judges were asked to provide ratings on all concepts using a five-point scale (1–5 stars). Given that the perception of high-level, abstract concepts is decidedly subjective, we collected 20 distinct judgments for each image.[9] The mean of judgments for an image was used as its final annotation. Participating judges were required to have an overall task acceptance rate greater than 98% and a total number of completed tasks greater than 5000. Beyond these general quality measures, no other selection criterion was imposed to avoid introducing potential selection bias(es). Judges were allowed to familiarize themselves with the collection through a pre-task warmup stage that enabled them to view and rate 9 images randomly selected from the highest and lowest star-rating group for each concept. For the main task, judges were provided 3 min to annotate each image. The presentation order was randomized for each judge to negate the effects of potential anchoring and presentation biases. Plots for mean vs. standard deviations for averaged judgments are available in Fig. 6.

[9] A total of 13,120 annotations (656 images × 20 annotations/image = 13,120 annotations) were collected. The reward for each annotation was $0.05.

Fig. 5. Data (label) distributions for images used in this work.

While the models presented in this work are designed to output a discrete, ordinal rank on a 7-point star-rating scale (2.0–5.0 with intervals of 0.5), the data collected from judges is obtained on a five-point scale, as five-point scales were found to be easier for judges to understand. In this case, averaging multiple judgments for an image can yield a real-valued score that does not match a valid star rating. Moreover, since judges only see a small part of the entire collection, it is unrealistic to expect a one-to-one correspondence between crowd-sourced judgments and star ratings. Therefore, instead of using a direct agreement measure between user judgments and predicted star ratings, we opt for measuring rank correlation — the ordinal association between predictions from the deep neural networks and averaged human judgments.

5.5. *Weak supervision from review text*

To examine the proposed approach for generating weak supervision from text, we analyzed 301,852 reviews comprising 1,625,100 constituent sentences. We computed sentence boundaries within a review using spaCy's statistical dependency parser (Montani et al., 2022). The survey questions, which these review sentences were eventually matched with, were selected from peer-reviewed research papers on

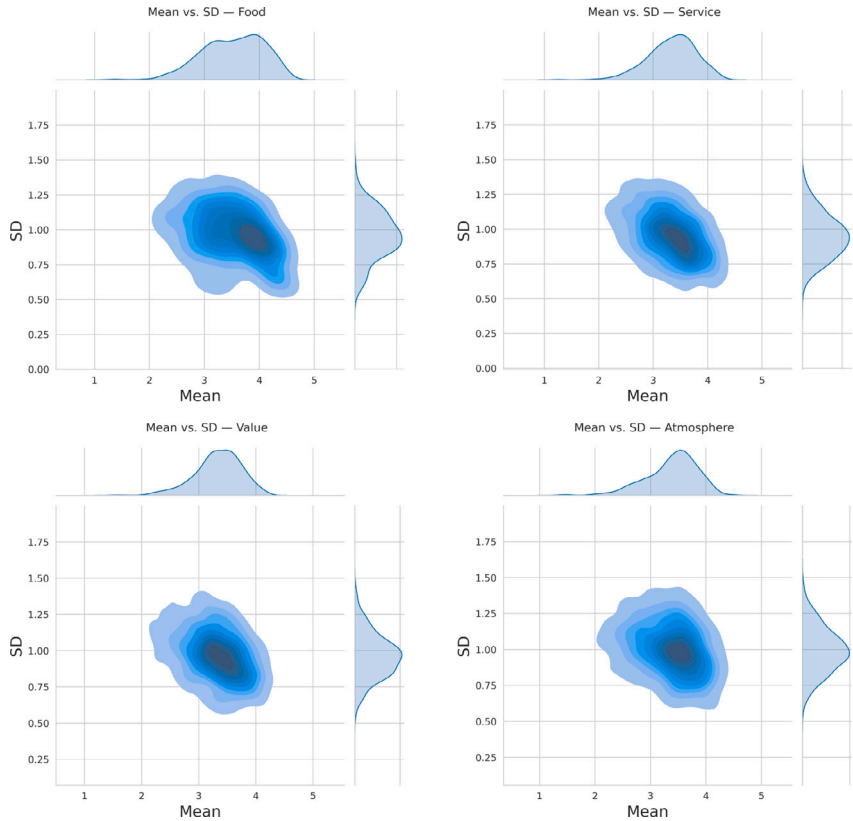

Fig. 6. Mean vs. Standard deviation density plots for human judgements.

restaurant and hospitality research and are provided in Appendix A. We computed the vector representation for sentences and survey questions once and saved them to disk. Additionally, we marked a small set of reviews where we obtained star ratings at the review level and retained these individually associated star ratings as a means for evaluation. Finally, we computed the inner product between sentence vectors and survey-question vectors as detailed in Section 4.

In Section 4.2, we examined how the noisy nature of review sentences can complicate the process of matching them with a survey question. To alleviate this problem, we match each review sentence with the top-6 most relevant survey questions. The sentiment of the review sentence is then attributed to these 6 survey questions. For all other survey questions, the detected sentiment is set to neutral (zero). Much like the choice of K in the K-nearest neighbors classification, this value was empirically chosen. When averaging sentiment on a specific concept over multiple sentences, we only average over non-zero (non-neutral) sentiment to avoid an observed problem with simple averaging of sentiment. Consider the example demonstrated in Fig. 4 – a simple averaging operation for aggregating sentiment over 4 sentences would sum up the sentiment on a concept in every sentence and divide it by 4 (the number of sentences). Since a concept is only expected to be discussed in a small subset of sentences (e.g., consumers will only talk about food in a small part of the review), a large majority of sentences will have zero sentiment (neutral or undetected sentiment). In the case of simple averaging, the sparsity would essentially dilute components of the vector corresponding to concepts present only in a small part of the review. To remedy this, we only average over non-zero (positive or negative) sentiment. At a high level, this design choice means we consider consumer sentiment on concepts that are not detected in a review sentence to be neutral.

5.5.1. *Evaluating concept detection*

The concept detection process, which involves associating a review sentence with an appropriate concept, is a crucial subtask within the label generation process. The performance of our proposed approach on this subtask has a significant impact on the overall quality of weak labels extracted from text.

To evaluate the performance of our approach on this subtask, we utilized training data from *SemEval 2014 - Task 4* as a test dataset to analyze its ability to capture the correct concept category. Specifically, we tested whether a concept known to be expressed in a sentence was also captured by our proposed approach. Within SemEval 2014 - Task 4, this corresponds to *Subtask 3: Aspect category detection*.[10] The dataset consists of 2581 sentences, each discussing one or more aspects out of five possible aspects (*food*, *service*, *price*, *ambiance*, and *anecdotes/miscellaneous*). Since our proposed approach does not capture miscellaneous discussions, we removed sentences with the 'anecdotes/miscellaneous' aspect from the dataset. Additionally, we replaced 'ambiance' with 'atmosphere' and 'price' with 'value' in the dataset to ensure consistent terminology. For each sentence, we computed the similarity with all survey questions. We then averaged similarities for survey questions associated under the same concept to generate a composite similarity score for that concept. The concept with the highest composite similarity score is considered as the detected concept. Since each sentence within the SemEval 2014 dataset can be associated with multiple aspects/concepts, we treated this as a multiclass-multilabel classification task and reported top-1 accuracy, precision, recall, and F1 scores.

5.5.2. *Evaluating weak label generation*

To evaluate our approach to generating sentiment vectors containing concept-specific sentiment scores, we utilize a small subset of user reviews that include ratings for the concepts studied in this work (food, service, value, and atmosphere). We employ a supervised BERT-based model trained on aspect-based sentiment analysis to obtain

[10] https://alt.qcri.org/semeval2014/task4/

sentiment predictions for review aspects. This enables us to compare the performance of our model with that of a state-of-the-art baseline model.

The baseline BERT model outputs discrete sentiment scores for each aspect, ranging from −1 (negative) to 0 (neutral) to +1 (positive). In contrast, our approach generates continuous sentiment scores on a scale of [−1, 1], where −1 represents extremely negative sentiment, 0 represents neutral sentiment, and 1 represents extremely positive sentiment. Across all label sources considered, the minimum value on this scale corresponds to highly negative sentiment, the midpoint corresponds to neutral sentiment, and the maximum value corresponds to highly positive sentiment.

To assess the effectiveness of our approach, we calculate the Spearman rank correlation between user feedback (in the form of star ratings) and the output from our proposed model. This correlation metric indicates the extent to which our approach captures the variation in user ratings. To provide context for interpreting these scores, we also compute the Spearman correlation between user ratings on restaurant concepts under study and the sentiment predictions generated by the supervised BERT-based model on these concepts.

5.6. Visualizing explanations from grad-CAM

Regarding the use of Grad-CAM in this work, we make two key assumptions. First, we assume that the rank of interest is the predicted rank; that is, if the model predicts a star rating of 4.5, this is the rating we aim to explain. Second, we assume that the convolutional layer used for weight computation is the final convolutional layer in the network.

Since the ordinal regression approach proposed in Diaz and Marathe (2019) distributes probability mass over adjacent categories (star ratings) (cf. Fig. 5), the probability mass for a specific rating is not entirely concentrated on that rating alone. In practice, this leads to fuzzier heatmaps with relatively large noise. To mitigate this issue, we employ an augmentation strategy where instead of obtaining a single Grad-CAM heatmap for the original image, we obtain multiple heatmaps from several augmented versions of the original image, reverse the original augmentation and average the de-augmented maps to produce a final heatmap. Specifically, we apply horizontal flipping and scaling operations as augmentations. In our experiments, this procedure helped suppress noise and led to generally sharper heatmaps.

6. Results and analysis

In this section, we seek to experimentally answer the following questions:

1. Is it possible to train a ranking model that can learn the relationship between images and high-level, abstract concepts using a weakly-supervised learning setup? Additionally, can gradient-based localization techniques provide visual explanations for predictions from such models?
2. Is it possible to extract information about high-level, abstract concepts from review text and does it line up with consumer feedback and results from contemporary state-of-the-art supervised approaches for detecting such concepts in text?

Below, we present the results of our experiments and provide an analysis of the findings.

6.1. Evaluating the performance of star-rating prediction models

Results from training the ordinal regression and classification models on the rating prediction tasks are presented in Table 3. For both classes of ordinal regression models, OR-EBC and OR-STE, scores on ordinal regression metrics are similar, indicating that most contemporary ordinal regression techniques have similar performance. Moreover, the usage of OR-STE does not come at the expense of performance. Furthermore, the mean absolute error on the test set for ordinal regression models is well below 0.5, which represents the inter-class separation between star ratings. The squared error measures, such as MSE and RMSE, which can indicate the presence of large outliers, are also reasonably small, indicating that misclassifications are not extremely large. In comparison to the classification baseline CLS-CE, the ordinal regression models consistently demonstrate the value of utilizing the information encoded in the separation between classes. Qualitative examples of predicted star ratings are provided in Fig. 9.

Our proposed model can generate both discrete and continuous-valued ranking information for images, which is particularly useful for practitioners who have access to an unannotated collection of images and need to determine which images score high on a certain attribute. The ability to generate continuous-valued ranking information is especially valuable in our case, as the limited label diversity significantly restricts the model's ability to predict lower ratings.

In Fig. 7, we observe that the imbalanced label set results in a restricted set of output classes. However, the model output expressed as an expectation over the output classes provides a more fine-grained measure that can be useful for ranking. This is demonstrated using Tripadvisor's example, where each restaurant has a large gallery of images but only a single aggregate score on a concept such as *Atmosphere*. To simulate a common use case for social media managers – ranking images from a single restaurant on a specific attribute – we pass all visual content from a restaurant through a model trained on the *Atmosphere* attribute. For a single Michelin Guide restaurant, *De Silveren Spiegel*, we computed the expected values of the output distributions for all images. We then selected 12 images at equal intervals, and these visual results are presented in Fig. 10. While Tripadvisor only provides a single aggregate star rating of *4.5* for food from this restaurant, the expected value of the output distribution indicates a large variety in ratings.

In the top-rated images, elements of gastronomic cuisine are visible, where a small quantity of food is placed in the center of a large, decorative plate. In this case, we surmise that the model has picked up elements of visual styles associated with such cuisine, since gastronomy restaurants are generally well-received for their food. Furthermore, the model can effectively pick up technical changes in lighting and presentation to differentiate between images with almost similar content. Thus, even for images labeled with the same star rating, a more fine-grained distinction can be made using the weighted average of the classifier output. The resulting real-valued scalar can be used to further rank visual content on a fine-grained level.

While the prediction error plots in Fig. 7 show that most outputs lean towards the dominant rating (4.0), the expected value of model outputs can track more subtle variations and displays a stronger correlation with human judgments, as shown in Fig. 8. Consequently, the *OR-STE* class of models can successfully provide both discrete (star-ratings) and fine-grained continuous (expected-value of the output distribution) measures for ranking content on a given attribute of interest.

Table 4 presents Spearman rank-correlations between the expected value of model outputs and judgments collected from our large-scale user study. The highest rank correlation is observed for "food", followed by "service", "value", and "atmosphere". To fully contextualize these results, it may be helpful to review certain attributes of the image dataset used in this work and of the high-level concepts under study.

The Tripadvisor dataset used in this work consists of images from restaurants and is predominantly composed of food-centric images. Thus, trained models learn to reason about high-level concepts like atmosphere or value through a largely food-centric image collection. In contrast, human judges tend to rate images in a one-sided way, where images containing visual elements that confirm their beliefs are rated higher than other images. In our crowd-sourcing experiments, human judges consistently tend to rate images containing explicitly visible atmosphere-related elements (furniture, paintings, etc.) or value-related elements (large quantities of food) higher than other images,

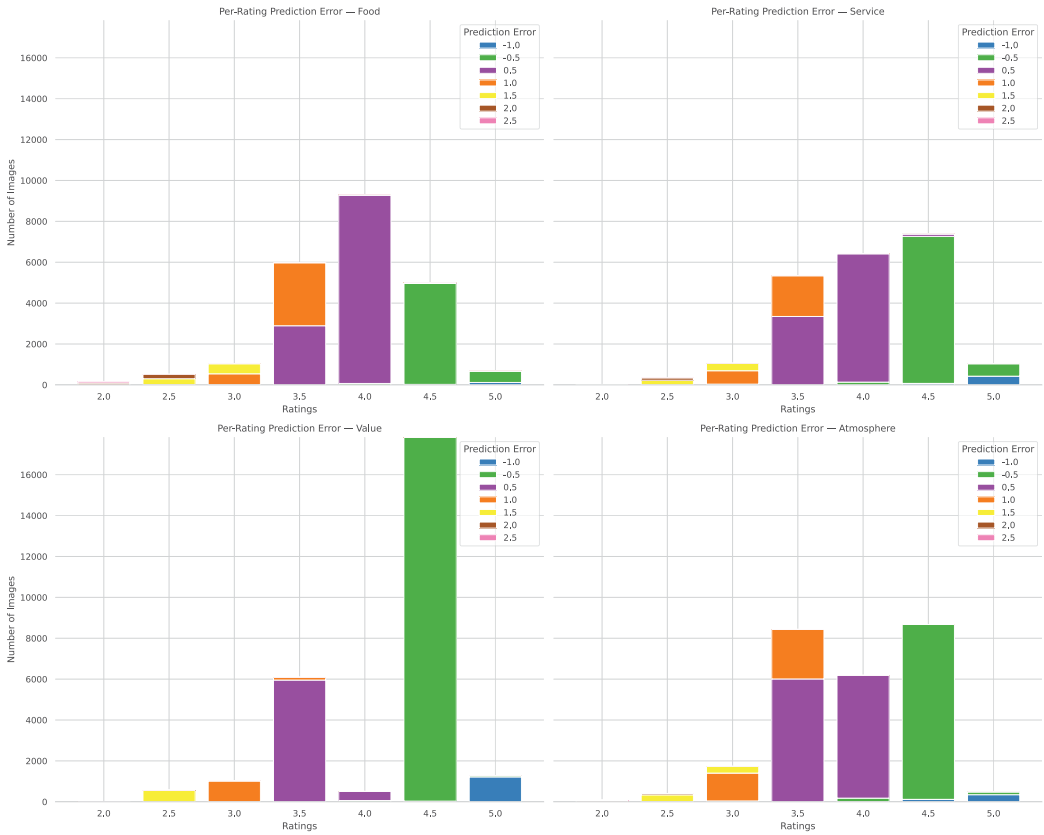

Fig. 7. Error-plots indicating the prediction error (Δ) for each rating.

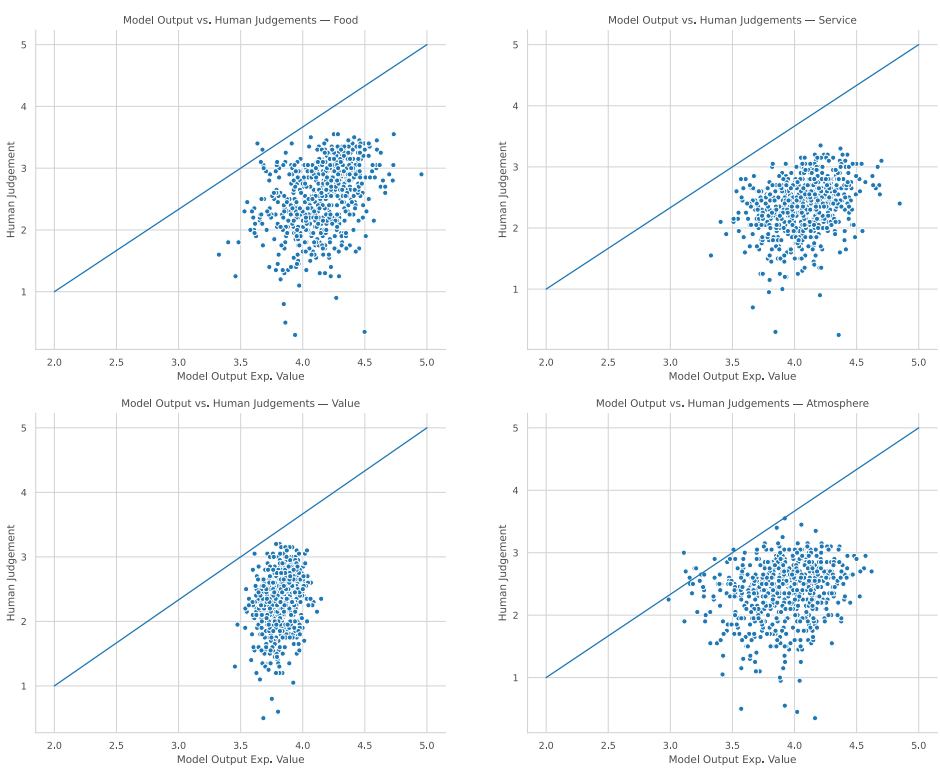

Fig. 8. Scatterplots indicating the relationship between the expected value of model outputs (x-axis) and human judgments (y-axis). The line indicates equivalence points between the human judgment scale (1–5 stars; the final rating is an average of 20 distinct judgments obtained on a five-point star-rating scale) and the range of expected values (2–5 stars).

Table 3
Evaluation Scores for Star-Rating prediction using the Ordinal Regression with Extended Binary Classification (OR-EBC), Ordinal Regression with Soft Target Encoding (OR-STE) and Classification with Categorical Cross-Entropy (CLS-CE) tasks. All scores are computed on the held-out test set. For all measures, lower is better.

Model	OR-EBC			OR-STE			CLS-CE		
Score / Attribute	MAE	MSE	RMSE	MAE	MSE	RMSE	MAE	MSE	RMSE
Food	0.301	0.218	0.467	0.303	0.234	0.484	0.352	0.265	0.515
Service	0.328	0.221	0.481	0.342	0.256	0.506	0.402	0.319	0.565
Value	0.337	0.232	0.470	0.342	0.227	0.476	0.423	0.333	0.577
Atmosphere	0.321	0.249	0.499	0.333	0.266	0.515	0.372	0.301	0.548

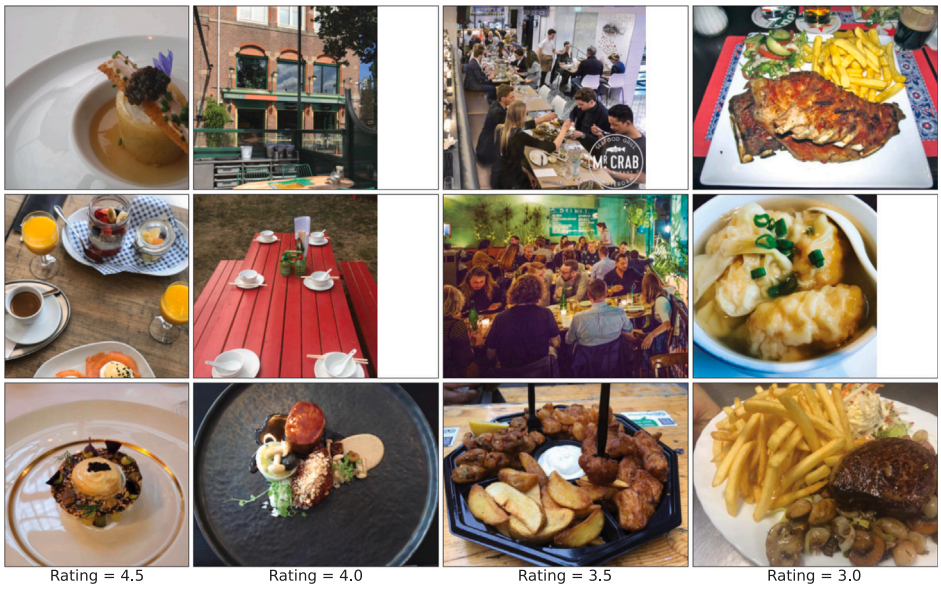

Fig. 9. Example outputs from the model trained for predicting the "Atmosphere" user-preference rating using the OR-STE task formulation. Predicted star ratings are on the bottom and decrease from left to right.

Fig. 10. Images from a single restaurant ranked using the expected value of the model outputs. The model is trained to predict the "Food Quality" star rating using the OR-STE task formulation. The scalar value under an image denotes the expected value of the predicted distribution over the star ratings.

Table 4
Spearman rank-correlation scores between the expected value of model outputs and the averaged human judgments. All correlations are significant ($p < 0.01$).

Attribute	Spearman Corr. with Human Judgements
Food	0.398
Service	0.330
Value	0.229
Atmosphere	0.208

as it seemingly confirms their beliefs about what constitutes a good atmosphere or value.

Consequently, it can be argued that in the case of complex concepts, human judgments can rely on external knowledge that is not entirely captured in the image. Information about food and service quality is abundantly available to human judges and the trained model. However, for elements like value and atmosphere that either require extra information (e.g., price information for ascertaining value) or rely on a specific type of images that are not abundantly available (e.g., pictures of the interior of the restaurant for judging the atmosphere), there is a low correlation between rankings obtained from human judges and those from the model.

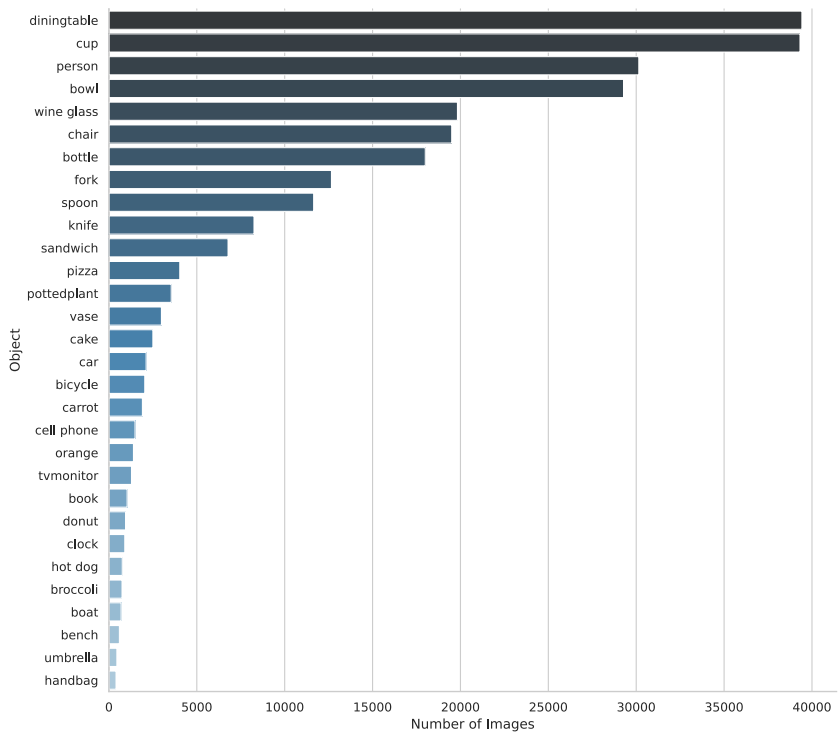

Fig. 11. Plot of object detection counts for the 30 most frequently detected objects in our dataset.

Previous works like Nanne et al. (2020) have examined visual UGC from an object-centric viewpoint. To illustrate why object-centric approaches may not apply to our case, we ran the YOLOv4 object detector on our image dataset (Bochkovskiy, Wang, & Liao, 2020). YOLOv4 can detect 80 unique objects in images. Object detection counts for the 30 most-frequent objects in our visual UGC dataset are plotted in Fig. 11. It can be seen that despite the relatively large size of the image dataset, the most frequently occurring objects were commonly expected articles in restaurant images such as cutlery (spoons, forks, knives, etc.), furniture (chairs, dining table, etc.), and food items (sandwich, pizza, cake, etc.). Unsurprisingly, visual UGC exclusively from a specific industry (in our case, restaurants) can exhibit a characteristic *narrow-domain* effect – a limited and predictive variability in all relevant aspects of the image's appearance (Smeulders et al., 2000). While object-centric analysis can be useful for data sources like Instagram that have high content diversity, similar treatments of narrow-domain sources like restaurant and hotel UGC may be hindered by the lack of object diversity.

6.2. Grad-CAM visualizations

In Fig. 12, we illustrate some Grad-CAM visualizations for ORD-STE models trained on the concepts "Food" and "Atmosphere" using the final bottleneck layer (conv5) in Resnet-50. Grad-CAM provides local explanations by highlighting regions of the image that support a specific prediction. From the first three images, it is evident that the model can weakly localize food items within the image, even though it was not explicitly trained to do so. In most predictions concerning food, the heatmap is roughly centered on the food item.

In the fourth image within Fig. 12, the heatmap highlights a part of the well-decorated table and the chandelier. Although this heatmap appears limited, it is important to note that many elements of atmospherics, such as lighting and textures, cannot be highlighted via Grad-CAM. Finally, a word of caution: while these outputs are often useful for debugging models, they do not indicate causal relationships.

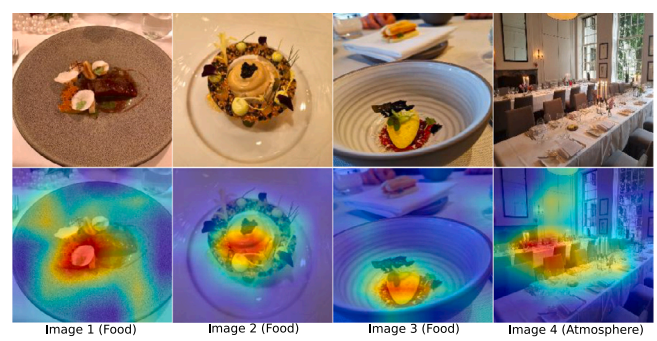

Fig. 12. Grad-Cam outputs for a ResNet-50 model trained to predict "Food" (Images 1–3) and "Atmosphere" (Image 4) star-ratings. Red and yellow regions within the image are *highly-activated* whereas blue and green regions have low activations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6.3. Validating weak-supervision mining from review text

The ability to correctly ascertain the concept or aspect discussed in a sentence is a critical component of the label mining process described in Section 4. Since each sentence within the SemEval 2014 - Task 4 can be associated with more than one aspect, we treat this as a multi-label classification task. We present scores for micro-, macro-, and weighted-average precision, recall, and F1 scores on the "aspect category detection" task in Table 5. Our results show that for most high-level, abstract concepts, precision is relatively high, while recall is lower. The high precision can be attributed to the precise framing of survey questions, which allows them to be unambiguously matched with review sentences that specifically provide a direct answer to the relevant question. In contrast, the low recall can be explained by the fact that complex sentences (e.g., sentences containing food names) cannot always be matched to the relatively limited set of survey queries. One way to improve the recall of this system is to expand

Table 5
Evaluation Scores on the "Aspect Category Detection" task.

	precision	recall	f1-score	support
food	0.968	0.345	0.509	1233
service	0.927	0.212	0.346	597
price	0.232	0.943	0.373	319
ambience	0.922	0.384	0.542	432
micro avg	0.497	0.395	0.440	2581
macro avg	0.762	0.471	0.442	2581
weighted avg	0.860	0.395	0.460	2581
samples avg	0.497	0.434	0.454	2581

the number of questions for each concept. Additionally, our experiments suggest that adding trivial reformulations of existing questions (e.g., "Was the food delicious", "Was the taste of the food great") to the pool of questions can also enhance the system's recall.

The supervised baseline used for benchmarking our model outputs sentiment on a given aspect through a ternary-valued output in $\{-1, 0, +1\}$, where -1 represents negative, 0 represents neutral, and $+1$ represents positive sentiments. In contrast, our proposed approach for extracting sentiment on an arbitrary attribute of interest outputs the detected sentiment through a linear, real-valued response in the range $[-1, 1]$, where -1 indicates extremely negative, 0 indicates neutral, and 1 indicates extremely positive sentiments. Across all label sources under consideration, the scale minimum consistently indicates highly negative sentiment, the midpoint indicates neutral sentiment, and the scale maximum indicates highly positive sentiment. The Spearman rank correlation between user feedback (in the form of star ratings) and the output from our proposed approach is a measure of how well our approach captures the variation in star ratings. To provide context for interpreting these scores, we also compute the Spearman correlation between user ratings on restaurant attributes and the sentiment detected by the supervised BERT-based model. The Spearman rank-correlation scores are provided in Table 6.

We use the results from the state-of-the-art BERT model as a benchmark to represent the performance of supervised techniques on this task. Our analysis reveals that, in most cases, the sentiment extracted via our method is moderately correlated with the star ratings provided by consumers, indicating that our proposed unsupervised approach performs reasonably well compared to state-of-the-art supervised approaches. While the supervised counterpart unsurprisingly outperforms our approach, our proposed unsupervised method offers greater flexibility and can be improved by expanding the set of questions used to determine the concept discussed in a sentence. This is an important development, as our proposed approach for extracting aspects related to sentences is entirely unsupervised and only requires practitioners to specify a set of questions related to the high-level, abstract concepts for which consumer sentiment needs to be mined. Furthermore, since our approach decomposes the task into separate aspect detection and sentiment analysis sub-tasks, it can benefit from independent technological advancements in question–answer retrieval and sentiment analysis research. Any increase in the performance of these sub-tasks should generally improve our proposed approach.

7. Discussion

In this section, we provide a concise overview of the practical implications of our work, highlighting its potential benefits for practitioners. We then discuss some limitations of our approach and identify promising avenues for future research. We begin by providing overarching propositions that place our results in the broader context. These are as follows:

- The central premise of our work is that it is possible to train a ranking model that can learn the relationship between images and high-level, abstract concepts using a weakly-supervised learning setup. This is a critical finding, as it demonstrates that it is possible to train models to predict ordinal ratings for high-level, abstract concepts entirely using weak supervision. A core operational assumption of our work is the reliance on central entities (in our case, restaurants) for generating supervision. In this framework, media is associated with a central entity, and the entity is associated with a concept. This allows for the rapid labeling of images with any concept that the entity has been associated with. However, such an approach is not limited to restaurants and can be applied to any domain where entities can be identified. For instance, in the case of user-centric platform like Instagram, each image can be linked to a distinct user, and the user can be associated with a concept. This allows for the rapid labeling of images with any concept that the user has been associated with.
- To validate our approach, we conducted a large-scale user study to collect human judgments on images. We found that the expected value of model outputs displays a moderate correlation with human judgments. Interestingly, the highest rank correlation was observed for the "food" attribute, followed by "service", "value", and "atmosphere"., indicating that the *concreteness effect* may be at play. This effect suggests that the more concrete an attribute is, the more likely it is to be rated consistently by human judges. This is an important finding, as it suggests that the model can effectively predict ratings for concrete attributes, such as food and service, but may struggle with more abstract attributes, such as value and atmosphere.
- Our results demonstrate that gradient-based localization techniques can provide visual explanations for predictions from ordinal regression models trained using the soft-label based ordinal regression framework presented in Section 3.2. In most cases, the model could reliably highlight parts of the image affecting the prediction.
- For the task of extracting information about high-level, abstract concepts from review text, our proposed approach can generate sentiment scores that are moderately correlated with user ratings. This is a significant finding, as it demonstrates that it is possible to extract sentiment on an arbitrary attribute of interest from review text without explicit labeling. Our approach is entirely unsupervised and only requires practitioners to specify a set of questions related to the high-level, abstract concepts for which consumer sentiment needs to be mined. This is a powerful tool for marketers, as it allows them to extract information on any attribute for which they can source appropriate questions.

These propositions provide a concise overview of the practical implications of our work and highlight its potential benefits for practitioners. We now discuss the contributions of our work to the literature, implications for practitioners, and limitations and future research directions.

7.1. Contributions to literature

Our work offers significant contributions to the literature on detecting high-level, abstract concepts in visual User-Generated Content (UGC), addressing limitations noted in contemporary works such as Liu et al. (2020). Specifically, our approach tackles two major issues: the inability to interpret model predictions and handle arbitrary concepts without available labels. From an algorithmic viewpoint, studies such as Liu et al. (2020) approach perceptual concept detection as a binary task, utilizing binary classifiers to determine if an attribute appears in an image. However, this binary perspective often falls short, as abstract attributes are perceived differently among individuals. Agreement between annotators regarding the strength of these concepts tends to be low, highlighting the subjective nature of high-level abstract

Table 6
Spearman rank-correlation scores between our proposed approach and the ground truth. Also provided are correlation scores between a strong supervised BERT baseline and the ground truth.

Attribute	Spearman Corr. b/w Ground-Truth and BERT Reference Model	Spearman Corr. b/w Ground-Truth and Proposed Approach	Number of Samples
Atmosphere	0.421	0.333	31,558
Food	0.476	0.427	83,240
Service	0.596	0.473	85,382
Value	0.534	0.418	84,426

concepts. Recent research in computer science recognizes this subjectivity, viewing labels as noisy consensuses rather than absolute truths. While a binary classification approach offers simplicity in evaluation, its utility for practitioners is limited. Our proposed method uses a ranking-based approach with two key advantages. First, the task formulation and network architecture resemble those of a classification task, facilitating training without significant structural changes and enabling the application of existing interpretability techniques such as Grad-CAM. Second, the task yields a distribution over ordinal labels, providing a fine-grained score for ranking images through expected values. Essentially, our model not only predicts star ratings but also offers a fine-grained scoring system useful for ranking images while maintaining the structural simplicity of a classification model.

While previous works have often attempted to construct such models within traditional supervised learning frameworks, a fundamental aspect of our approach is the reliance on weak labels based on the understanding that manual labeling efforts will never suffice to annotate entire real-world UGC collections, given the rapid growth rate of visual UGC. Weak labeling offers a method for operationalizing UGC by leveraging existing sources of supervision. Moreover, our decision to utilize entities for generating supervision can be applied across various use cases. While restaurants serve as the central entities in our research, similar core entities can be identified in other sources of visual UGC, such as Instagram, where each image can be linked to a distinct user. Incorporating entities enables swift labeling of images with any concept that the entity has been associated with.

Our proposed label mining approach, which extracts labels from text, is a powerful tool for practitioners seeking to examine UGC on attributes that the associated restaurant has not been annotated for. With the help of a small set of questions, marketers can rapidly extract weak labels indicating consumer satisfaction (or dissatisfaction) from review text. This is a powerful tool for marketers as it allows them to extract information on any attribute for which they can source appropriate questions. Furthermore, this process is entirely unsupervised and extremely fast.

Finally, visualizations obtained with Grad-CAM illustrate that deep neural networks can often provide valuable insights for practitioners. These visualizations can serve two powerful roles. First, they alleviate concerns arising from the "black-box" nature of such models by providing visual explanations for a prediction. Social media managers can use these insights to identify visual elements associated with favorable consumer perception and ensure that they are sufficiently promoted in firm-generated content and advertising for the restaurant. Second, such visualizations allow practitioners to diagnose convolutional neural networks for potential weaknesses. For example, if a model trained to predict the rating of an attribute fails to localize visual properties known by practitioners to be relevant, such models may be capturing erroneous correlations and might need to be fixed before deployment.

7.2. Implications for practitioners

For digital marketers and social media managers, this model provides a powerful information management tool for ranking visual content in terms of high-level, abstract concepts valued by consumers. For instance, the proposed model can recommend images that would be better received on a concept, allowing restaurants to select and promote this content through their digital marketing platform. They can also take steps to improve the visibility and discoverability of such elements within their establishment. Since our model relies on a computational technique, it is faster than manual curation.

Similarly, large digital platforms like TripAdvisor often need to select a subset of images to showcase a restaurant. This model provides a robust method for selecting content that clearly communicates abstract concepts important to consumers. Moreover, these models help tackle the "cold-start" problem, where new restaurants may not have enough user feedback for a rating. In such instances, automated visual content analysis allows platforms to assign temporary star ratings until sufficient user feedback is obtained.

The label mining process proposed in Section 4 allows practitioners to generate labels with minimal effort, leveraging the well-established neural question–answer similarity framework. Unlike approaches that rely on a curated set of seed words to bootstrap the aspect detection task (in our work, the task of detecting if a review sentence discusses food, service, value, or atmosphere) and require extensive collection efforts, which can be brittle and susceptible to typos and variations in terms, our approach aligns the message conveyed in review sentences with the broader semantics of the "query" sentences. This alignment provides relative resilience against variations in expression, a frequent challenge in user-generated text characterized by abundant typos, alternative word spellings, and diverse phrasing choices.

7.3. Limitations and future research

Since our work relies on User-Generated Content (UGC) as the primary visual content source, it can be argued that our models learn how consumers, business owners, and even professional photographers choose to portray high-level attributes rather than how they manifest within a restaurant. This limitation presents an information gap: since we can only see a restaurant through UGC, our model can only learn elements that were deemed critical to the service and posted online. Furthermore, our models rely exclusively on weak and noisy labels, which, despite their relative ease of obtaining, remain a relatively noisy source of supervision. Additionally, our proposed approach relies on a set of central entities (restaurants in our study) to which every image can be tied. This may not be the case with every UGC collection. In the absence of a means for associating a restaurant with an image, labels cannot be propagated and our weak-labeling approach does not work. In such cases, practitioners may have to seek an entirely different source of supervision. Moreover, since information about unsuccessful restaurants is not archived by Tripadvisor, our work is subject to natural limitations arising from survival bias.

There are also some technical limitations with our unsupervised approach for mining supervision signals from review text. For instance, consider a scenario where a user tries to forcefully discover aspects that do not exist in the review texts. Since our proposed label-mining approach assumes that each review sentence can be associated with some survey question(s), it will associate the sentence with the *most* meaningful questions without warning the user that their choice of aspects is generally meaningless. The development of a system that can inform the user that no signals can be mined for a specific aspect from a set of review texts remains a goal for future research.

Table A.1
List of questions associated with relevant concepts.

Concept	Question	Source
Food	Were you satisfied with the quality of your food?	MacLaurin and MacLaurin (2000)
	Was the taste of food excellent?	MacLaurin and MacLaurin (2000)
	Was the food served at the proper temperature?	Gupta, McLaughlin, and Gomez (2007)
	Was the appearance of food excellent?	MacLaurin and MacLaurin (2000)
	Was the quantity of food sufficient?	MacLaurin and MacLaurin (2000)
	Was the quality of food consistent?	MacLaurin and MacLaurin (2000)
Service	Was the service good?	Gupta et al. (2007)
	Did the restaurant serve food exactly as you ordered?	MacLaurin and MacLaurin (2000)
	Did the restaurant provide quick and prompt service?	MacLaurin and MacLaurin (2000)
	Was the staff able and willing to give information?	MacLaurin and MacLaurin (2000)
	Did the restaurant have well-trained, competent and experienced employees?	MacLaurin and MacLaurin (2000)
	Was the staff attentive?	Gupta et al. (2007)
	Was the staff polite and friendly?	Gupta et al. (2007)
	Did the staff seat you at your table quickly?	Gupta et al. (2007)
	Was food served in a timely-manner?	Gupta et al. (2007)
	Was the server's appearance neat and clean?	Gupta et al. (2007)
Value	Did the restaurant provide value-for-the-money?	MacLaurin and MacLaurin (2000)
	Was the total cost appropriate for the food and service you received?	Gupta et al. (2007)
	Were the menu prices too high?	Gupta et al. (2007)
	Were you pleased with the amount of food you were served?	Gupta et al. (2007)
Atmosphere	Was the atmosphere great?	Authors
	Did the decor suit the theme?	MacLaurin and MacLaurin (2000)
	Was the dining area comfortable?	MacLaurin and MacLaurin (2000)
	Was the staff properly dressed?	MacLaurin and MacLaurin (2000)
	Did the restaurant have a comfortable dining area?	Authors
	Did the restaurant have comfortable seats?	Authors

8. Conclusion

Visual content is here to stay, and its online presence will continue to grow exponentially in an increasingly digital world. This burgeoning data source will decisively shape consumer beliefs, expectations, and perceptions. However, analyzing this vast amount of data remains an onerous task. Beyond the obvious challenges posed by scale and content diversity, the visual higher-order constructs that interest consumers and marketers alike are often complex and dynamic. Our work presents several methodological advances that enable practitioners to examine large-scale User-Generated Content (UGC) automatically and categorize images according to how consumers perceive them in terms of high-level business-relevant higher-order constructs. By doing so, we alleviate the burden of manually scanning and analyzing large visual content collections and provide marketers with a powerful tool for filtering through this deluge of data. While deep learning approaches have revolutionized visual content analysis, their applicability is still limited by their reliance on labeled data. Our work demonstrates how existing sources of information, despite their noisy or imperfect nature, can provide a powerful supervisory signal for training predictive models. Moreover, in the absence of such data, we show that text mining approaches coupled with traditional marketing tools like survey questionnaires can be used to mine labels from rich, unstructured sources of information like review text. Furthermore, neural networks can also serve as a source of insights for marketers. Modern gradient-based visualization techniques can provide marketers with visual explanations for model predictions, allowing them to understand existing deep learning models and improve their existing visual content. Therefore, our work presents advances that not only enable practitioners to rapidly train models but also provide a means for interpreting and learning from such models.

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CRediT authorship contribution statement

Ujjwal Sharma: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stevan Rudinac:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Joris Demmers:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Willemijn van Dolen:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Marcel Worring:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. List of survey questions

The set of questions used for the supervision mining approach presented in Section 4 are listed in Table A.1.

Appendix B. ResNet-50

Fig. B.1 provides a detailed model schematic for the ResNet-50 model, the deep neural network that serves as the visual inference backbone for our work.

1. Blocks with a dashed boundary are "bottleneck" building blocks and consist of multiple convolutional layers as demonstrated in the expanded view.
2. Due to space constraints, each layer has only been shown with 3 bottleneck blocks. The actual number of bottleneck blocks in each layer is provided in square brackets under the layer name.

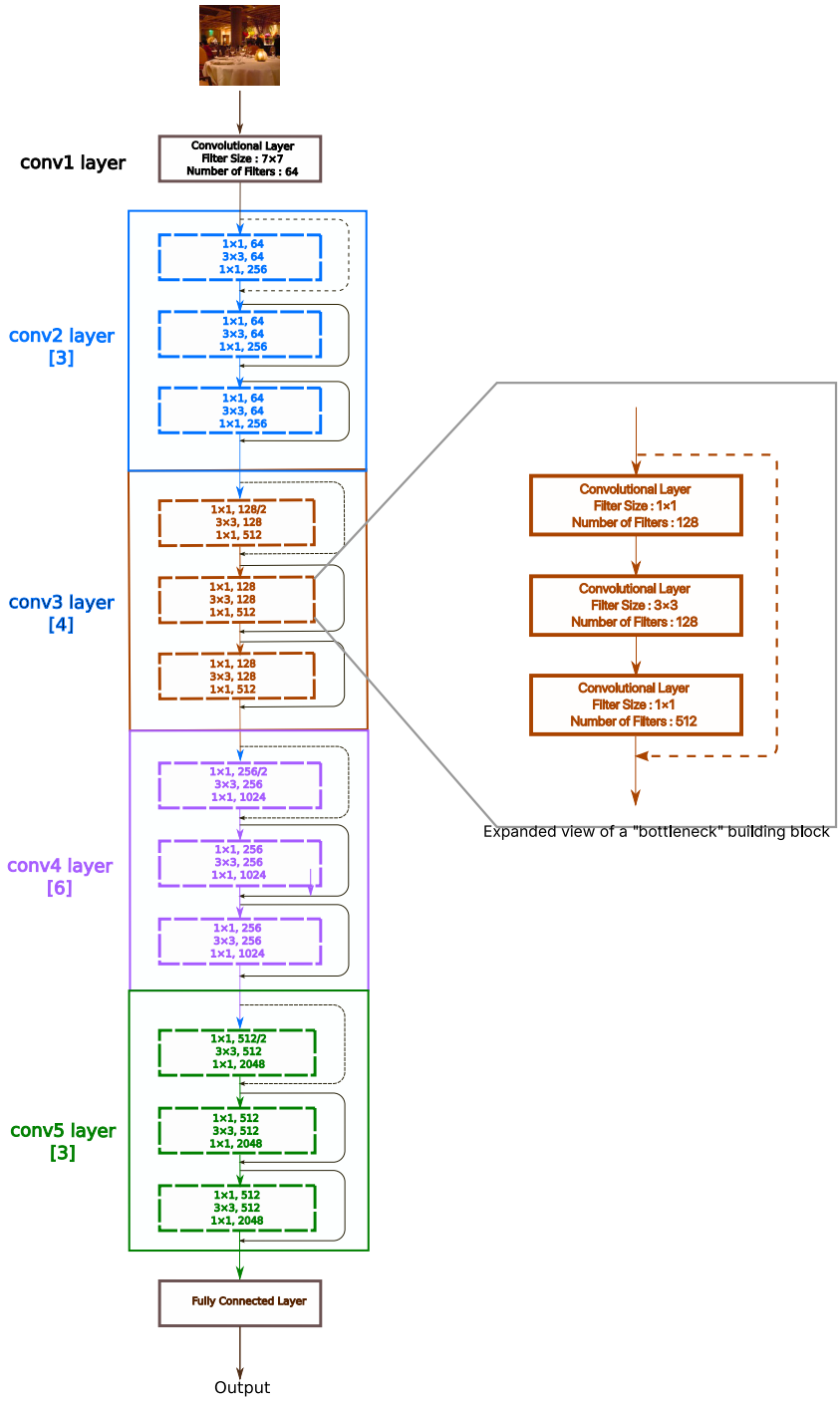

Fig. B.1. An illustration of the ResNet-50 model.

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